SCOPE OF WORK:

REMOVE AND INSTALL HVAC EQUIPMENT, DUCTWORK AND PIPING. INSTALL EMERGENCY GENERATOR ON ROOF. WORK IN CONJUNCTION WITH ALT-CO M01244603-I1.

(TR-1) SPECIAL INSPECTIONS:

1. FIRE-RESISTANT PENETRATIONS AND JOINTS	BC 1705.17
2. MECHANICAL SYSTEMS	BC 1705.21
3. HEATING SYSTEMS	BC 1705.31
4. CHIMNEYS AND VENTS	BC 1705.32
5. EMERGENCY AND STANDBY POWER SYSTEMS(GENERATOR)	BC 1705.36
6. EXHAUST AND VENT SYSTEMS IN PLENUMS AND DUCTS	BC 1705.39

(TR-1) PROGRESS INSPECTIONS:

1. ENERGY CODE COMPLIANCE INSPECTIONS	BC 110.3.5
2. FINAL	BC 110.5

(TR-8) ENERGY CODE PROGRESS INSPECTIONS:

1. HVAC-R AND SERVICE WATER HEATING EQUIPMENT	(IB3), (IIB3)
2. HVAC-R AND SERVICE WATER HEATING CONTROLS	(IB4), (IIB4)
3. HVAC-R AND SERVICE WATER PIPING DESIGN AND INSULATION	(IB5), (IIB5)
4. MAINTENANCE INFORMATION	(ID1), (IID1)
5. SHUTOFF DAMPERS	(IIB2)
6. DUCT LEAKAGE TESTING, INSULATION AND DESIGN	(IB6), (IIB6)

GENERAL NOTES

1. ALL WORK SHALL CONFORM TO THE RULES AND REGULATIONS. CODES AND ORDINANCES OF FEDERAL, STATE AND LOCAL GOVERNMENTAL AGENCIES HAVING JURISDICTION OVER THE PROJECT, NO WORK SHOWN SPECIFIED OR IMPLIED SHALL BE CONSTRUCTED TO CONFLICT WITH ANY GOVERNMENT AGENCY OR BUILDING AUTHORITY MANDATE.
2. NO WORK SHALL BEGIN UNTIL A BUILDING PERMIT HAS BEEN ISSUED.
3. GENERAL CONTRACTOR SHALL PROVIDE GENERAL LIABILITY INSURANCE, WORKMEN COMPENSATION INSURANCE AND DISABILITY INSURANCE BEFORE PERFORMING ANY WORK IN THIS BUILDING.
4. THE CONTRACTOR SHALL INSPECT THE SITE AND CHECK AND VERIFY EXISTING AND PROPOSED CONDITIONS BEFORE STARTING ANY WORK. THE CONTRACTOR SHALL BRING THE DISCREPANCIES TO THE ARCHITECT'S OR ENGINEER'S ATTENTION BY WRITING PRIOR TO SUBMIT ANY QUOTES OR ESTIMATES PRIOR TO COMMENCING WORK.
5. ALL NEW WORK CONFORMS TO THE NEW YORK STATE ENERGY CODE.
6. ALL FEES, PERMITS APPLICATIONS, INSPECTIONS, CERTIFICATES AND SIGN-OFF SHALL BE SECURED BY THE CONTRACTOR AND PAID FOR BY THE OWNER UNLESS OTHERWISE NOTED.
7. ALL NOTES HEREIN MENTIONED ALONG WITH THOSE ON THE VARIOUS DRAWINGS SHALL APPLY TO ALL DRAWINGS AND IS PART OF THE CONTRACT.
8. THE CONTRACTOR SHALL PROVIDE ALL THE LABOR AND MATERIAL NECESSARY FOR THE COMPLETION OF THE WORK AS SHOWN ON THE DRAWINGS.
9. THE CONTRACTOR SHALL CARRY OUT ALL THE WORK IN A FIRST CLASS WORKMAN-LIKE MANNER. HE SHALL MAINTAIN A SAFE. CLEAN JOB SITE AT ALL TIMES, FREE FROM DEBRIS ACCUMULATION, AND SHALL LEAVE THE JOB SITE BROOM CLEAN AT THE COMPLETION OF THE PROJECT.
10. THE CONTRACTOR SHALL BE SOLELY RESPONSIBLE FOR ANY AND ALL DAMAGE TO NEW OR EXISTING CONDITIONS CAUSED BY HIMSELF, HIS WORKMEN OR HIS SUBCONTRACTOR. HE SHALL REPAIR SUCH DAMAGE AT NO ADDITIONAL COST TO THE OWNER UNLESS OTHERWISE NOTED IN WRITING PRIOR TO STARTING WORK.
11. ALL MATERIALS AND CONSTRUCTION TO BE INCORPORATED IN THE WORK SHALL BE IN STRICT ACCORDANCE WITH THE LATEST EDITION OF THE ASTM SPECIFICATIONS APPLICABLE AND TO CONFORM TO THE STANDARDS AND RECOMMENDATIONS OF THE VARIOUS TRADE INSTITUTES (ACI, AISC, ETC.) WHERE APPLICABLE. ALL MATERIALS INCORPORATED INTO THE WORK SHALL BE NEW AND FREE FROM DEFECTS UNLESS OTHERWISE NOTED.
12. CONTRACTORS SHALL BE RESPONSIBLE FOR ADEQUATELY BRACING AND PROTECTING ALL WORK DURING CONSTRUCTION AGAINST DAMAGE, BREAKAGE, COLLAPSE, DISTORTIONS, STANDARDS AND GOOD PRACTICE. CONTRACTOR AGREES THAT HE SHALL ASSUME SOLE AND COMPLETE RESPONSIBILITY FOR THE SITE CONDITIONS DURING THE COURSE OF CONSTRUCTION OF THIS PROPERTY, THAT REQUIREMENT SHALL APPLY CONTINUOUSLY AND NOT BE LIMITED TO NORMAL WORKING HOUR THE CONTRACTOR SHALL DEFEND, INDEMNIFY & HOLD THE ARCHITECT OR ENGINEER HARMLESS FROM ANY ALL LIABILITY, REAL OR ALLEGED IN CONNECTION WITH THE PERFORMANCE OF WORK ON THIS PROJECT.
13. THE ARCHITECT OR ENGINEER OF DESIGN HAS NOT BEEN RETAINED FOR ANY FILED SUPERVISION OR INSPECTION. HIS RESPONSIBILITY IS LIMITED TO THE ACCURACY OF THE PLANS.
14. MATERIALS NOT SPECIFIED BY THEIR MAKE, STYLE, COLORS ETC. SHALL BE APPROVED BY THE OWNER PRIOR TO PURCHASING AND INSTALLATION BY CONTRACTORS.
15. THIS PLAN IS APPROVED ONLY FOR WORK INDICATED ON THE APPLICATION SPECIFICATION SHEET. ALL OTHER MATTERS SHOWN ARE NOT TO BE RELIED UPON, OR TO BE CONSIDERED AS EITHER-BEING APPROVED OR IN ACCORDANCE WITH APPLICABLE CODES.

BUILDING DEPARTMENT NOTES:

1. ALL CONSTRUCTION SHALL CONFORM TO THE REQUIREMENTS OF THE NEW YORK CITY ADMINISTRATIVE BUILDING CODE EFFECTIVE 2022 AND AMENDMENTS THERETO.
2. GENERAL CONTRACTOR SHALL OBTAIN GENERAL LIABILITY INSURANCE, WORKMEN'S COMPENSATION INSURANCE AND DISABILITY INSURANCE BEFORE PERFORMING ANY WORK.
3. ALL NEW WORK SHALL CONFORM TO THE NEW YORK CITY ENERGY CODE.
4. ALL FEES, PERMITS, APPLICATIONS, INSPECTIONS CERTIFICATION, SIGN-OFFS SHALL BE SECURED AND PAID FOR BY THE CONTRACTOR.
5. THE GENERAL CONTRACTOR SHALL PROVIDE ALL THE LABOR AND MATERIAL FOR THE COMPLETION OF THE WORK AS SHOWN ON THE DRAWINGS AND SPECIFICATIONS.
6. SAFETY OF PUBLIC AND PROPERTY DURING CONSTRUCTION OPERATIONS SHALL BE IN ACCORDANCE WITH ARTICLE 1B.
7. THE WORK SHOWN ON THE DRAWINGS DO NOT INCLUDE CHANGE OF EGRESS, USE OR OCCUPANCY

MECHANICAL CODE NOTES:

1. ENERGY CODE COMPLIANCE NOTE: TO THE BEST OF MY KNOWLEDGE, BELIEF, AND PROFESSIONAL JUDGMENT, THE PLANS AND SPECIFICATIONS ARE IN COMPLIANCE WITH THE 2020 NYC ECC
2. PERMITS: CONTRACTOR SHALL FILE FOR AND OBTAIN PERMITS FOR ALL MECHANICAL SYSTEMS. PERMITS SHALL COMPLY WITH 2022 MECHANICAL CODE, SECTION MC105 AND WITH 2022 ADMINISTRATIVE CODE, TITLE 28 – ARTICLE 105.
3. LISTING AND LABELING OF APPLIANCES: ALL APPLIANCES REGULATED BY THE 2022 MECHANICAL CODE SHALL BE LISTED AND LABELED AS REQUIRED UNDER MC301.7 AND MC301.9
4. VIBRATION ISOLATION: ALL MECHANICAL EQUIPMENT AND APPLIANCES SHALL BE PROVIDED WITH SUPPORTED AND RESTRAINT AS REQUIRED UNDER 2022 MECHANICAL CODE SECTION 313
5. FIRE STOPPING COMPLIANCE: FIRE STOPPING AT CONSTRUCTION PENETRATIONS SHALL COMPLY WITH 2022 MECHANICAL CODE SECTION MC602.5 AND 2014 BUILDING CODE SECTION BC 714
6. DUCT SYSTEMS: DUCT SYSTEMS USED FOR THE MOVEMENT OF AIR IN HVAC SYSTEMS SHALL COMPLY WITH 2022 MECHANICAL CODE CHAPTER 6. DUCT INSULATION SHALL MEET THE REQUIREMENTS OF MC 604 AND THE ENERGY CONSERVATION CONSTRUCTION CODE OF NEW YORK STATE
7. MATERIALS: MATERIALS MUST BE USED, TESTED AND APPROVED FOR USE IN ACCORDANCE TO 2022 ADMINISTRATIVE CODE, ARTICLE 113.
8. VENTILATION: VENTILATION SHALL BE PROVIDED BY THE MEANS OF ONE OR MORE OF THE FOLLOWING: FANS, ERVS, TRICKLE VENTS, OR THE EQUIVALENT PER MC 401.2.

HVAC BUILDING DEPARTMENT NOTES:

1. A CONTROLLED INSPECTION AND TEST WILL BE CONDUCTED UNDER THE DIRECTION OF A LICENSED PROFESSIONAL ENGINEER OR ARCHITECT OR OTHER PERSON HAVING NOT LESS THAN (5) YEARS EXPERIENCE SUPERVISING THE INSTALLATION OF VENTILATING SYSTEMS. THE TEST WILL SHOW COMPLIANCE WITH THE BUILDING ADMINISTRATIVE CODE AND REFERENCE STANDARDS.
2. THE LICENSED PROFESSIONAL ENGINEER OR ARCHITECT OR OTHER PERSON NOT HAVING LESS THAN (5) YEARS EXPERIENCE SUPERVISING THE INSTALLATION OF VENTILATING SYSTEMS AND CONDUCTING SUCH TESTS WILL FILE A CERTIFICATE AND REPORT OF TEST THAT THE SYSTEM COMPLIES WITH THE APPLICABLE LAWS.
3. A STATEMENT WILL BE FILED BY THE OWNER (OR TENANT) IN POSSESSION THAT THE VENTILATING SYSTEM WILL BE KEPT IN CONTINUOUS OPERATION DURING NORMAL OCCUPANCY OF THE PREMISES.
4. ALL FIRE DAMPERS ARE TO BE OF TYPE APPROVED BY THE BOARD OF FIRE UNDERWRITERS. WHERE ENTERING OR LEAVING SHAFTS, FIRE DAMPERS ARE TO BE EQUIVALENT TO WALL RATING.
5. THE LATEST RULES OF BUILDING CODES ARE TO BE COMPLIED WITH.
6. SMOKE AND FIRE DETECTION SYSTEMS TO BE INSTALLED IN ACCORDANCE WITH THE BUILDING ADMINISTRATIVE CODE AND REFERENCE STANDARDS.

CERTIFICATE OF COMPLIANCE BY CONTRACTOR:

1. CERTIFICATE OF COMPLIANCE SHALL BE OBTAINED BY CONTRACTOR IN ACCORDANCE WITH THE ADMINISTRATIVE CODE.
2. M.E.A. AND B.S. & A. NUMBERS FOR EQUIPMENT SHALL BE SUBMITTED AS REQUIRED BY THE BUILDING CODE.
3. CERTIFICATE OF COMPLIANCE SHALL BE REQUIRED FOR THE USE AND OPERATION OF SERVICE EQUIPMENT INCLUDING BUT NOT LIMITED TO:
   a. AIR CONDITIONING EQUIPMENT
   b. VENTILATION SYSTEMS

SERVICE EQUIPMENT PERMIT:

AS PER 2022 BUILDING CODE SECTION BC106.2, SERVICE EQUIPMENT PERMITS ARE REQUIRED FOR THE INSTALLATION OR ALTERNATION OF SERVICE EQUIPMENT, INCLUDING BUT NOT LIMITED TO AIR CONDITIONING AND VENTILATION SYSTEMS.

SEISMIC BRACING, RESTRAINTS & SUPPORT:

SUBJECT TO BUILDINGS WITH A SEISMIC DESIGN IN CATEGORY C, ALL MECHANICAL COMPONENTS WITH Ip=1.0 ARE EXEMPT FROM SEISMIC REQUIREMENTS. ALL MECHANICAL COMPONENTS WITH Ip=1.5 SHALL BE DESIGNED AND CONSTRUCTED TO SEISMIC REQUIREMENTS.

ASBESTOS NOTE:

THE CLIENT SHALL HAVE THE DUTY TO INFORM THE ENGINEER OF THE PRESENCE OF ASBESTOS AND OTHER TOXIC SUBSTANCES THAT ARE KNOWN OR ARE OF A REASONABLE CONCERN. IF ASBESTOS OR OTHER TOXIC SUBSTANCES ARE DISCOVERED BY THE ENGINEER TO BE PRESENT OR HAVE A LIKELIHOOD OF BEING PRESENT DURING WORK DONE TO THE PROJECT, HE SHALL, IN HIS OWN DISCRETION, HAVE THE RIGHT TO SUSPEND WORK ON THE PROJECT. THE CLIENT SHALL PROMPTLY OBTAIN A QUALIFIED EXPERT TO SAFELY REMOVE OR SUPERVISE THE REMOVAL OF SUCH ASBESTOS OR TOXIC SUBSTANCES. THE CLIENT SHALL INDEMNIFY ENGINEERS, AGENTS, AND EMPLOYEES FROM ALL LIABILITY, INCLUDING BUT NOT LIMITED TO HARM CAUSED PEOPLE AND THE COST OF LEGAL FEES THAT RESULT FROM EXPOSURE TO ASBESTOS AND OTHER TOXIC SUBSTANCES.

BUILDING CLASSIFICATION STATEMENT (MC 106.4):

WHERE APPLICABLE TO THE PROPOSED WORK, THE STATEMENT SHALL IDENTIFY:

1. THE OCCUPANCY GROUP OR GROUPS THAT APPLY TO THE PARTS OF THE BUILDING IN ACCORDANCE WITH SECTION 302 OF THE NEW YORK CITY BUILDING CODE;
2. THE OCCUPANCY GROUP OF THE MAIN USE OR DOMINANT OCCUPANCY OF THE BUILDING;
3. THE CONSTRUCTION CLASS OF THE BUILDING IN ACCORDANCE WITH SECTION 602 OF THE NEW YORK CITY BUILDING CODE;
4. THE STRUCTURAL OCCUPANCY/RISK CATEGORY IN ACCORDANCE WITH TABLE 1604.5 OF THE NEW YORK CITY BUILDING CODE;
5. THE HEIGHT OF THE BUILDING AS DEFINED IN SECTION 502.1 OF THE NEW YORK BUILDING CODE;
6. THE APPLICABLE MEASUREMENTS TO THE HIGHEST AND LOWEST LEVEL OF THE FIRE DEPARTMENT ACCESS; AND
7. WHETHER THE BUILDING IS INSIDE OR OUTSIDE OF THE FIRE DISTRICTS.

MECHANICAL ABBREVIATIONS

A	AMPERES	KX	KITCHEN EXHAUST
AC	AIR CONDITIONER	L	LENGTH
ACC	AIR COOLED CONDENSER	LAT	LEAVING AIR TEMPERATURE
ACS	AUTOMATIC CONTROL SYSTEM	LBS	POUNDS
ACU	AIR CONDITIONING UNIT	LDB	LEAVING DRY BULB
AD	ACCESS DOOR	LF	LINEAR FEET
AFF	ABOVE FINISHED FLOOR	LWB	LEAVING WET BULB
AHU	AIR HANDLING UNIT	LWT	LEAVING WATER TEMPERATURE
AL	ALUMINUM	MAX.	MAXIMUM
AP	ACCESS PANEL	MB	MIXING BOX
BF	BOOSTER FAN	MBH	THOUSAND BTU PER HOUR
BG	BOTTOM OF GRILLE	MER	MECHANICAL EQUIPMENT ROOM
BHP	BRAKE HORSE POWER	MHP	MOTOR HORSE POWER
BMS	BUILDING MANAGEMENT SYSTEM	MIN.	MINIMUM
BR	BOTTOM OF REGISTER	MM	MILLIMETER
BT	BOTTOM OF THROAT	NC	NORMALLY CLOSED
BTU	BRITISH THERMAL UNIT	NO	NORMALLY OPEN
BTU/H	BTU PER HOUR	NIC	NOT IN CONTRACT
CC	COOLING COIL	NO.	NUMBER
CD	CEILING DIFFUSER	NTS	NOT TO SCALE
CFM	CUBIC FEET PER MIN	OA	OUTSIDE AIR
CV	CONSTANT VOLUME	OAI	OUTSIDE AIR INTAKE
CVR	CONSTANT VOLUME REGULATOR	OD	OUTSIDE DIAMETER
CP	CONDENSATE PUMP	PD	PRESSURE DROP
D	DEPTH	PF	PREFILTER
DIA.	DIAMETER	PRV	PRESSURE REDUCING VALVE
DN	DOWN	PSI	POUNDS PER SQUARE INCH
DWG	DRAWING	PSIA	ABSOLUTE PSI
EA	EXHAUST AIR	PSIG	GAUGE PSI
EAT	ENTERING AIR TEMPERATURE	R	RISE
EDB	ENTERING DRY BULB TEMPERATURE	RA	RETURN AIR
EX.	EXISTING	RH	RELATIVE HUMIDITY
EL.	ELEVATION	RHC	REHEAT COIL
ELEC.	ELECTRICAL	RM	ROOM
EQ	EQUAVALENT	RPM	REVOLUTIONS PER MINUTE
EWB	ENTERING WET BULB TEMPERATURE	SA	SUPPLY AIR
EXH	EXHAUST	SF	SUPPLY FAN
EXP	EXPANSION	SLD	STRIPLINE DIFFUSER
°F	DEGREES FAHRENHEIT	SP	STATIC PRESSURE
FA	FREE AREA (FT^2)	SPEC.	SPECIFICATION
FC	FLEXIBLE CONNECTION	SS	STAINLESS STEEL
FD	FIRE DAMPER	T	TEMPERATURE
FG	FINISHED GRADE	TF	TERMINAL FILTER
FF	FINISHED FLOOR	TG	TOP OF GRILL
FLA	FULL LOAD AMPERES	TR	TOP OF REGISTER
FO	FUEL OIL	TRD	TRANSFER DUCT
FOR	FUEL OIL RETURN	TRF	TRANSFER FAN
FOS	FUEL OIL SUPPLY	TRG	TRANSFER GRILL
FPM	FEET PER MINUTE	TV	TURNING VANE
FPI	FINS PER INCH	TX	TOILET EXHAUST
FPS	FEET PER SECOND	TYP.	TYPICAL
FR	FLOOR REGISTER	UON	UNLESS OTHERWISE NOTED
FT	FEET	V	VOLTS
FTR	FINNED TUBE RADIATION	W	WIDTH
GA.	GAUGE	W/	WITH
GAL.	GALLON	W/O	WITHOUT
GPM	GALLONS PER MINUTE	WB	WET BULB
HC	HEATING COIL	WC	WATER COLUMN
HR	HOUR	WG	WATER GAUGE
HUM.	HUMIDIFIER	WMS	WIRE MESH SCREEN
HZ	HERTZ	WP	WORKING PRESSURE
ID	INNER DIAMETER	WSA	WIRE SIZE AMPS
IN.	INCH/INCHES	VA	VENTILATION AIR
KW	KILOWATT		

PROFESSIONAL STATEMENT

THE PROJECT REGISTERED DESIGN PROFESSIONAL IN CHARGE WILL PERFORM REVIEWS/QUALITY CHECKS FOR THE BUILDING RELATING TO ENERGY COMPLIANCE. A REQUIRED STATEMENT FROM THE ARCHITECT (OR MULTIPLE STATEMENTS FROM THE DESIGNERS, ARCHITECT & ENGINEERS) WILL BE SUBMITTED, SHOWING THAT THE ITEM(S) UNDER THEIR RESPONSIBILITY WERE REVIEWED FOR ENERGY COMPLIANCE BY THE BUILDING INSPECTOR. SUCH REVIEWS BUILDING INSPECTOR FOR THEIR REVIEW. ALL REQUIRED ENERGY STATEMENTS WILL BE LISTED IN THE PLANS. NARRATIVES MUST BE SUBMITTED AS A DOCUMENT IN THE SUBMITTAL PACKAGE REFERENCING THE APPROPRIATE DRAWING.

NYC BUILDING DEPARTMENT APPROVAL NOTE

THIS PLAN SHALL BE APPROVED ONLY FOR WORK INDICATED ON THE SPECIFICATION SHEET. ALL OTHER WORK MATTERS. ALL UNSPECIFIED WORK SHALL NOT BE CONSIDERED AS APPROVED OR IN ACCORDANCE WITH APPLICABLE CODES

FLOOD ZONE STATEMENT

PROPERTY IS NOT IN A SPECIAL FLOOD HAZARD AREA (SFHA)

ENERGY CODE ANALYSIS STATEMENT

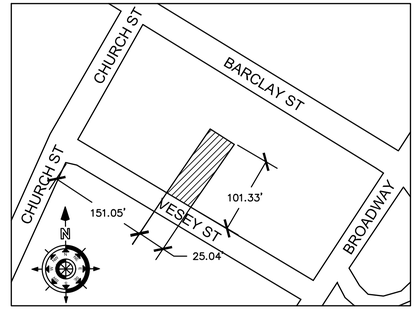
TO THE BEST OF MY KNOWLEDGE, BELIEF AND PROFESSIONAL JUDGEMENT, ALL WORK DONE UNDER THIS APPLICATION IS COMPLIANT TO THE 2020 NEW YORK CITY ENERGY CONSERVATION CODE

DRAWING LIST

NO.	DWG NO.	DWG NAME	S1
1	M001.00	MECHANICAL NOTES, SYMBOLS, DRAWING LIST, AND SITE PLAN	●
2	M051.00	EXISTING SECOND AND THIRD FLOOR PLAN	●
3	M052.00	EXISTING FOURTH AND FIFTH FLOOR PLAN	●
4	M053.00	EXISTING ROOF	●
5	M101.00	PROPOSED SUB-CELLAR AND CELLAR FLOOR PLAN	●
6	M102.00	PROPOSED FIRST FLOOR PLAN	●
7	M103.00	PROPOSED SECOND AND THIRD FLOOR PLAN	●
8	M104.00	PROPOSED FOURTH AND FIFTH FLOOR PLAN	●
9	M105.00	PROPOSED ROOF AND BULKHEAD PLAN	●
10	M301.00	MECHANICAL AIR AND REFRIGERANT PIPE RISER DIAGRAM	●
11	M401.00	MECHANICAL EQUIPMENT SCHEDULE	●
12	M402.00	MECHANICAL EQUIPMENT SCHEDULE II	●
13	M501.00	MECHANICAL DETAILS	●
14	M502.00	MECHANICAL DETAILS II	●
15	M503.00	MECHANICAL DETAILS III	●
16	M504.00	MECHANICAL DETAILS IV	●
17	M505.00	MECHANICAL DETAILS V	●
18	M601.00	MECHANICAL SPECIFICATIONS	●

MECHANICAL SYMBOLS

- SINGLE LINE DUCTWORK – NEW
- SINGLE LINE DUCTWORK – EXISTING
- SINGLE LINE DUCTWORK – DEMO
- SINGLE LINE DUCTWORK W/ ACOUSTICAL LINING
- DUCT UNDER POSITIVE PRESSURE
- DUCT UNDER NEGATIVE PRESSURE (EXHAUST)
- DUCT UNDER NEGATIVE PRESSURE (RETURN)
- CEILING DIFFUSER GRILL
- CEILING EXHAUST GRILL
- CEILING RETURN GRILL / PLENUM RETURN
- (T) THERMOSTAT
- (TS) TEMPERATURE SENSOR
- (SW) SWITCH
- (CP) CONDENSATE PUMP
- POINT OF CONNECTION
- POINT OF DISCONNECTION
- FD/AD FIRE DAMPER W/ ACCESS DOOR
- FSD/AD FIRE/SMOKE DAMPER W/ ACCESS DOOR
- PSD/AD POST SMOKE DAMPER W/ ACCESS DOOR
- MD MOTORIZED DAMPER
- BD BACKDRAFT DAMPER
- VD VOLUME DAMPER
- (XX) CFM VALUE
- (N) NEW
- (E) EXISTING
- (R) RELOCATED
- (D) DEMOLISH



1 PLOT PLAN
N.T.S.

RELATED APPLICATIONS:	
ALT-CO GENERAL CONSTRUCTION:	DOBNOW#M01244603-I1
PLUMBING:	DOBNOW#M01244603-S2
SPRINKLER:	DOBNOW#M01244603-S3
FUEL STORAGE:	DOBNOW#M01244603-S4
BOILER:	DOBNOW#M01244603-S5

PROPERTY INFORMATION

18 VESEY STREET
NEW YORK, NY, 10007
BLOCK: 88
LOT: 4
BIN: 1001234
COMMUNITY BOARD: 101
ZONING: C5-3, LM (LOWER MANHATTAN)
ZONING MAP: 12B
LOT SIZE = 2,537 SQFT.

HEIGHT: 73 FT
STORIES: 5 STORIES
OCCUPANCY CLASSIFICATION: R-1
CONSTRUCTION CLASSIFICATION: 3
NON-FIREPROOF
MULTIPLE DWELLING CLASSIFICATION: HACB

CONSTRUCTION AREA
SQUARE FOOTAGE: 4175 SQFT

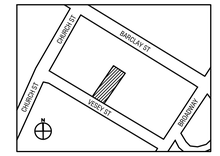
THIS BUILDING WILL BE VACANT DURING CONSTRUCTION

REVIEW REQUESTED UNDER 2022 MECHANICAL CODE

ALL CITY ENGINEERING, P.C.
33 Great Neck Road #3
Great Neck, NY 11021
(516)321-9490
info@allcityengineering.com

OWNER:
SEAFARERS INTERNATIONAL HOUSE
123 EAST 15TH STREET
NEW YORK, NY 1003
TEL: (212) 677-4800
EMAIL: LUCAS@SIHNYC.ORG

SITE PLAN



DoB STAMPS:

No.	DATE	REVISION
	01/08/26	BID SET

NOTE: Drawing may be printed at reduced scale

IT IS A VIOLATION OF THE STATE EDUCATION LAW SECTION 7209 (2) FOR ANY PERSON TO ALTER AN ITEM IN ANY WAY UNLESS SUCH PERSON IS ACTING UNDER THE DIRECTION OF A LICENSED PROFESSIONAL ENGINEER, AND THE ENGINEER STAMPS SUCH CHANGES

PROJECT:

18 VESEY STREET
MANHATTAN, NY
BLOCK: 88 LOT: 4

DRAWING TITLE:

MECHANICAL NOTES, SYMBOLS, DRAWING LIST, AND SITE PLAN

DATE:	01/06/2026
PROJECT No.:	25086
CHECKED BY:	
DRAWN BY:	PY/LZ/BL

SEAL AND SIGNATURE

DWG. No.:
M001.00

Sheets: 01 of 18

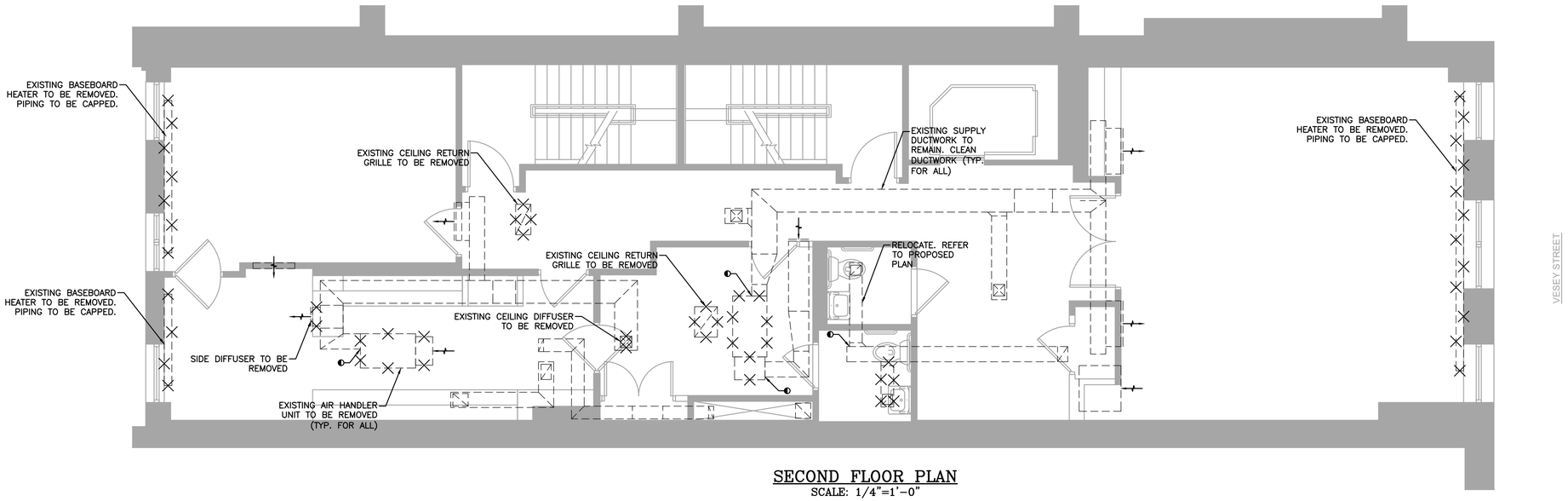
N.Y.C D.O.B No. :
M01244603-S1

DUCTWORK NOTES:

1. PROVIDE 18 GAGE ALUMINUM DUCT FOR CLOTH DRYER EXHAUST
2. PROVIDE 18 GA. DUCTWORK FOR RESIDENTIAL KITCHEN EXHAUST
3. PROVIDE MIN. 24 GAGE FOR GENERAL DUCTWORK
4. ALL INTAKE OPENING LOCATIONS SHALL COMPLY WITH 2022 MC SECTION 401.4
5. ALL EXHAUST OUTLET LOCATION SHALL COMPLY WITH 2022 MC SECTION 501.3.1
6. PROVIDE MIN. 1.5" INSULATION FOR ALL DUCTWORK DEDICATED FOR OA
7. PROVIDE MIN. 1.5" INSULATION FOR ALL CONCEALED DUCTWORK DEDICATED FOR AC SUPPLY
8. PROVIDE 1.5" ACOUSTICAL LINING FOR AC DUCTWORK
9. ALL INTERIOR AND EXTERIOR MECHANICAL EQUIPMENT SHALL COMPLY WITH 2022 MC SECTION 928
10. ALL DUCTWORK WITH A CROSS-SECTIONAL AREA OF 20 IN^2 OR LESS PASSING THROUGH FIRE-RATED ASSEMBLIES EXCEPT FIRE WALLS SHALL NOT REQUIRE FIRE DAMPERS OR FIRE SMOKE DAMPERS, AS PER 2022 MC 607.1.1 & 2022 BC SECTION 716.5
11. PROVIDE ACCESS DOOR FOR ALL FIRE DAMPER, FIRE/SMOKE DAMPER AND POST-FIRE SMOKE DAMPER
12. ALL OUTSIDE AIR INTAKE AND EXHAUST MOTORIZED DAMPERS SHALL BE CLASS I LEAKAGE TYPE
13. DUCT CONSTRUCTION AND INSTALLATION SHALL STRICTLY FOLLOW ANSI/SMACNA 006-2020 STANDARD.
14. PROVIDE VOLUME DAMPER/CORD OPERATED DAMPER FOR EVERY BRANCH OF DUCTWORK.
15. VERIFY IN FIELD IF EXISTING SUPPLY DUCTS HAVE VOLUME DAMPERS, IF NOT PROVIDE ONE.
16. VERIFY IN FIELD IF EXISTING DUCTWORK HAS FIRE DAMPER AND FIRE SMOKE DAMPER HAS BEEN INSTALL, IF NOT PROVIDE ONE.
17. VERIFY IN FIELD IF MOTORIZED/BACKDRAFT DAMPERS ARE INSTALLED. IF FAN/GOOSENECK SUPPLY OR EXHAUST IS 300 CFM OR MORE PROVIDE MOTORIZED DAMPER AND IF 300 CFM OR LESS PROVIDE BACKDRAFT DAMPER.

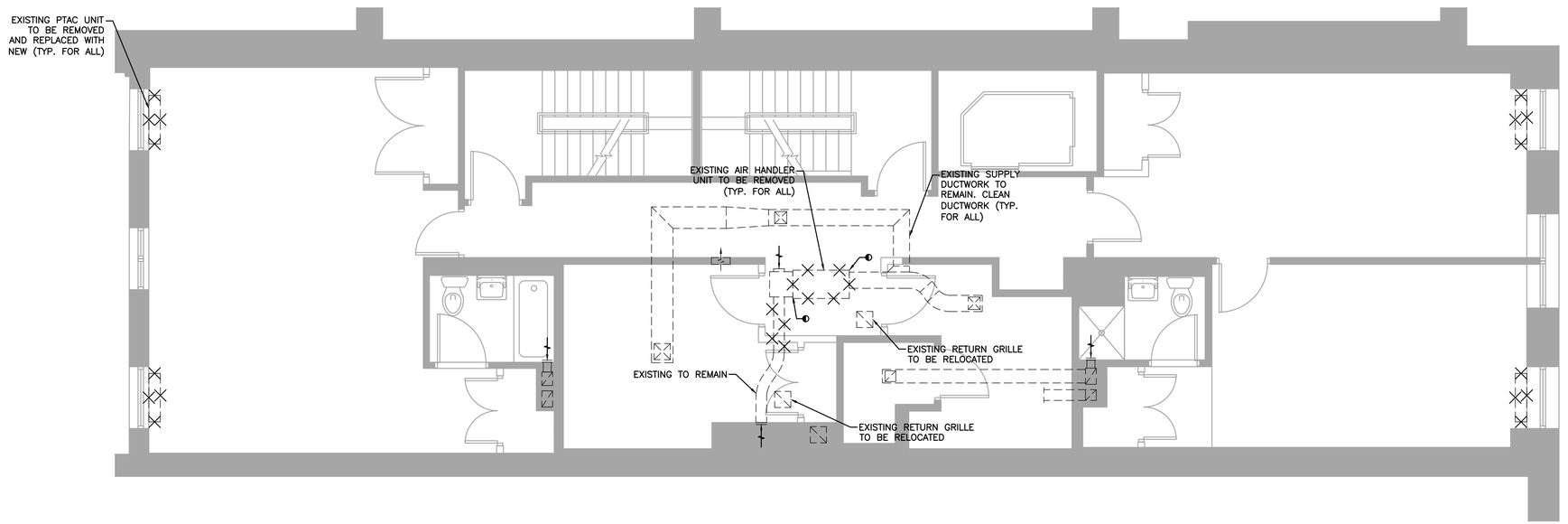
HVAC PIPING NOTES:

1. PROVIDE PROPER INSULATION THICKENESS FOR REFRIGERANT PIPING AS PER NYCECC TABLE C403.11.3 AND/OR 2016 ASHRAE TABLE 6.8.3.1
2. CONDENSATE PIPE MATERIAL AND SIZING SHALL COMPLY WITH 2022 MC SECTION 307.2.2.
3. PROVIDE THERMAL EXPANSION LOOP FOR REFRIGERANT PIPING AT EVERY 100FT OF RUN.
4. PROVIDE CONDENSATE PUMP FOR ALL AC UNITS (WITH THE EXCEPTION OF WALL MOUNTED UNITS AND HPAC UNITS) AND ROUTE CONDENSATE TO NEAREST INDIRECT DRAIN
5. PER ADDENDUM TO ASHRAE STANDARD 15-2024. VENTILATION FOR SHAFTS CONTAINING REFRIGERANT PIPING USING R-32 AND R-454B REFRIGERANT SHALL NOT BE REQUIRED WHERE ALL THE REFRIGERANT PIPE OR TUBE IS CONTINUOUS AND HAS BEEN TESTED IN ACCORDANCE WITH SECTION 9.13 OF THE SAFETY STANDARD FOR REFRIGERATION SYSTEMS AS WELL AS BEING TESTED AT 650 PSI OPERATING PRESSURE FOR 24 HOURS.



SECOND FLOOR PLAN

SCALE: 1/4"=1'-0"



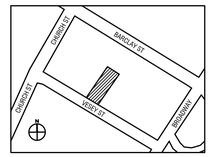
THIRD FLOOR PLAN

SCALE: 1/4"=1'-0"

ACE
ALL CITY ENGINEERING, P.C.
33 Great Neck Road #3
Great Neck, NY 11021
(516)321-9490
info@allcityengineering.com

OWNER:
SEAFARERS INTERNATIONAL HOUSE
123 EAST 15TH STREET
NEW YORK, NY 1003
TEL: (212) 677-4800
EMAIL: LUCAS@SIHNYC.ORG

SITE PLAN



DoB STAMPS:

No.	DATE	REVISION
	01/08/26	BID SET

NOTE: Drawing may be printed at reduced scale

IT IS A VIOLATION OF THE STATE EDUCATION LAW SECTION 7209 (2) FOR ANY PERSON TO ALTER AN ITEM IN ANY WAY UNLESS SUCH PERSON IS ACTING UNDER THE DIRECTION OF A LICENSED PROFESSIONAL ENGINEER, AND THE ENGINEER STAMPS SUCH CHANGES

PROJECT:

18 VESEY STREET
MANHATTAN, NY
BLOCK: 88 LOT: 4

DRAWING TITLE:

EXISTING SECOND AND THIRD FLOOR PLAN

DATE:	01/06/2026
PROJECT No.:	25086
CHECKED BY:	
DRAWN BY:	PY/LZ/BL

SEAL AND SIGNATURE

DWG. No.:
M051.00

Sheets: 02 of 18

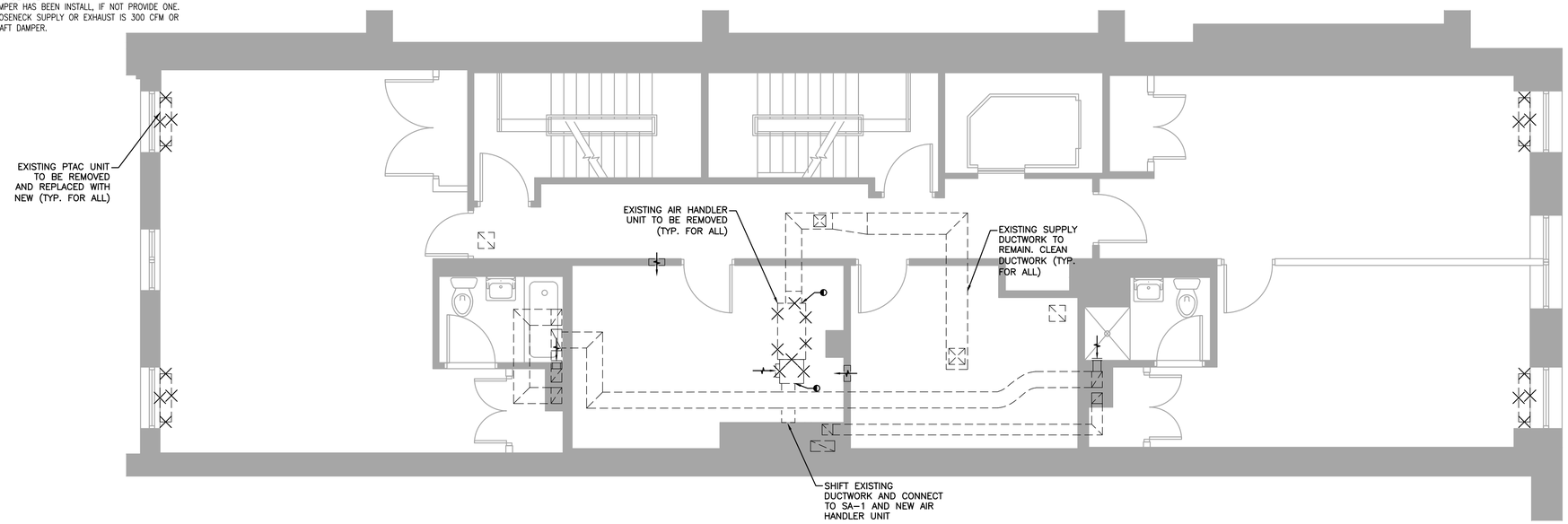
N.Y.C D.O.B No. :
M01244603-S1

DUCTWORK NOTES:

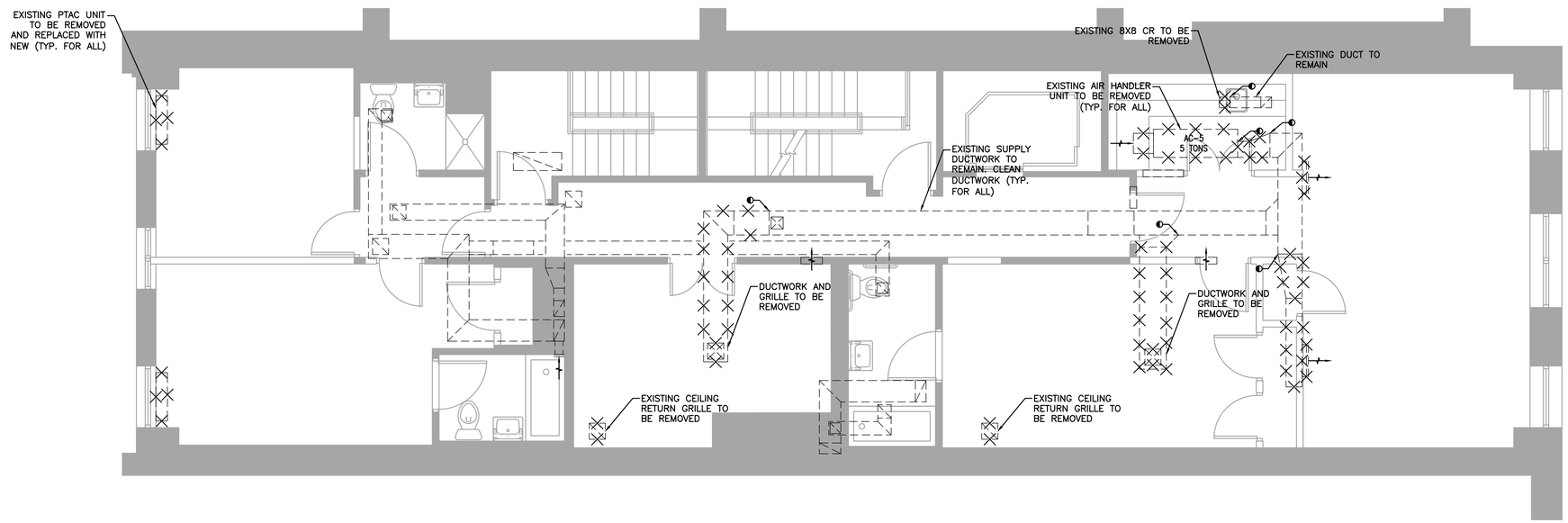
1. PROVIDE 18 GAGE ALUMINUM DUCT FOR CLOTH DRYER EXHAUST
2. PROVIDE 18 GA. DUCTWORK FOR RESIDENTIAL KITCHEN EXHAUST
3. PROVIDE MIN. 24 GAGE FOR GENERAL DUCTWORK
4. ALL INTAKE OPENING LOCATIONS SHALL COMPLY WITH 2022 MC SECTION 401.4
5. ALL EXHAUST OUTLET LOCATION SHALL COMPLY WITH 2022 MC SECTION 501.3.1
6. PROVIDE MIN. 1.5" INSULATION FOR ALL DUCTWORK DEDICATED FOR OA
7. PROVIDE MIN. 1.5" INSULATION FOR ALL CONCEALED DUCTWORK DEDICATED FOR AC SUPPLY
8. PROVIDE 1.5" ACOUSTICAL LINING FOR AC DUCTWORK
9. ALL INTERIOR AND EXTERIOR MECHANICAL EQUIPMENT SHALL COMPLY WITH 2022 MC SECTION 928
10. ALL DUCTWORK WITH A CROSS-SECTIONAL AREA OF 20 IN^2 OR LESS PASSING THROUGH FIRE-RATED ASSEMBLIES EXCEPT FIRE WALLS SHALL NOT REQUIRE FIRE DAMPERS OR FIRE SMOKE DAMPERS, AS PER 2022 MC 607.1.1 & 2022 BC SECTION 716.5
11. PROVIDE ACCESS DOOR FOR ALL FIRE DAMPER, FIRE/SMOKE DAMPER AND POST-FIRE SMOKE DAMPER
12. ALL OUTSIDE AIR INTAKE AND EXHAUST MOTORIZED DAMPERS SHALL BE CLASS I LEAKAGE TYPE
13. DUCT CONSTRUCTION AND INSTALLATION SHALL STRICTLY FOLLOW ANSI/SMACNA 006-2020 STANDARD.
14. PROVIDE VOLUME DAMPER/CORD OPERATED DAMPER FOR EVERY BRANCH OF DUCTWORK.
15. VERIFY IN FIELD IF EXISTING SUPPLY DUCTS HAVE VOLUME DAMPERS, IF NOT PROVIDE ONE.
16. VERIFY IN FIELD IF EXISTING DUCTWORK HAS FIRE DAMPER AND FIRE SMOKE DAMPER HAS BEEN INSTALL, IF NOT PROVIDE ONE.
17. VERIFY IN FIELD IF MOTORIZED/BACKDRAFT DAMPERS ARE INSTALLED. IF FAN/GOOSENECK SUPPLY OR EXHAUST IS 300 CFM OR MORE PROVIDE MOTORIZED DAMPER AND IF 300 CFM OR LESS PROVIDE BACKDRAFT DAMPER.

HVAC PIPING NOTES:

1. PROVIDE PROPER INSULATION THICKENESS FOR REFRIGERANT PIPING AS PER NYCECC TABLE C403.11.3 AND/OR 2016 ASHRAE TABLE 6.8.3.1
2. CONDENSATE PIPE MATERIAL AND SIZING SHALL COMPLY WITH 2022 MC SECTION 307.2.2.
3. PROVIDE THERMAL EXPANSION LOOP FOR REFRIGERANT PIPING AT EVERY 100FT OF RUN.
4. PROVIDE CONDENSATE PUMP FOR ALL AC UNITS (WITH THE EXCEPTION OF WALL MOUNTED UNITS AND HPAC UNITS) AND ROUTE CONDENSATE TO NEAREST INDIRECT DRAIN
5. PER ADDENDUM TO ASHRAE STANDARD 15-2024. VENTILATION FOR SHAFTS CONTAINING REFRIGERANT PIPING USING R-32 AND R-454B REFRIGERANT SHALL NOT BE REQUIRED WHERE ALL THE REFRIGERANT PIPE OR TUBE IS CONTINUOUS AND HAS BEEN TESTED IN ACCORDANCE WITH SECTION 9.13 OF THE SAFETY STANDARD FOR REFRIGERATION SYSTEMS AS WELL AS BEING TESTED AT 650 PSI OPERATING PRESSURE FOR 24 HOURS.



FOURTH FLOOR PLAN
SCALE: 1/4"=1'-0"

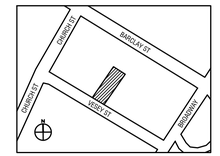


FIFTH FLOOR PLAN
SCALE: 1/4"=1'-0"

ACE
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SITE PLAN



DoB STAMPS:

No.	DATE	REVISION
	01/08/26	BID SET

NOTE: Drawing may be printed at reduced scale

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PROJECT:

18 VESEY STREET
MANHATTAN, NY
BLOCK: 88 LOT: 4

DRAWING TITLE:

EXISTING FOURTH AND FIFTH FLOOR PLAN

DATE:	01/06/2026
PROJECT No.:	25086
CHECKED BY:	
DRAWN BY:	PY/LZ/BL

SEAL AND SIGNATURE

DWG. No.:
M052.00

Sheets: 03 of 18

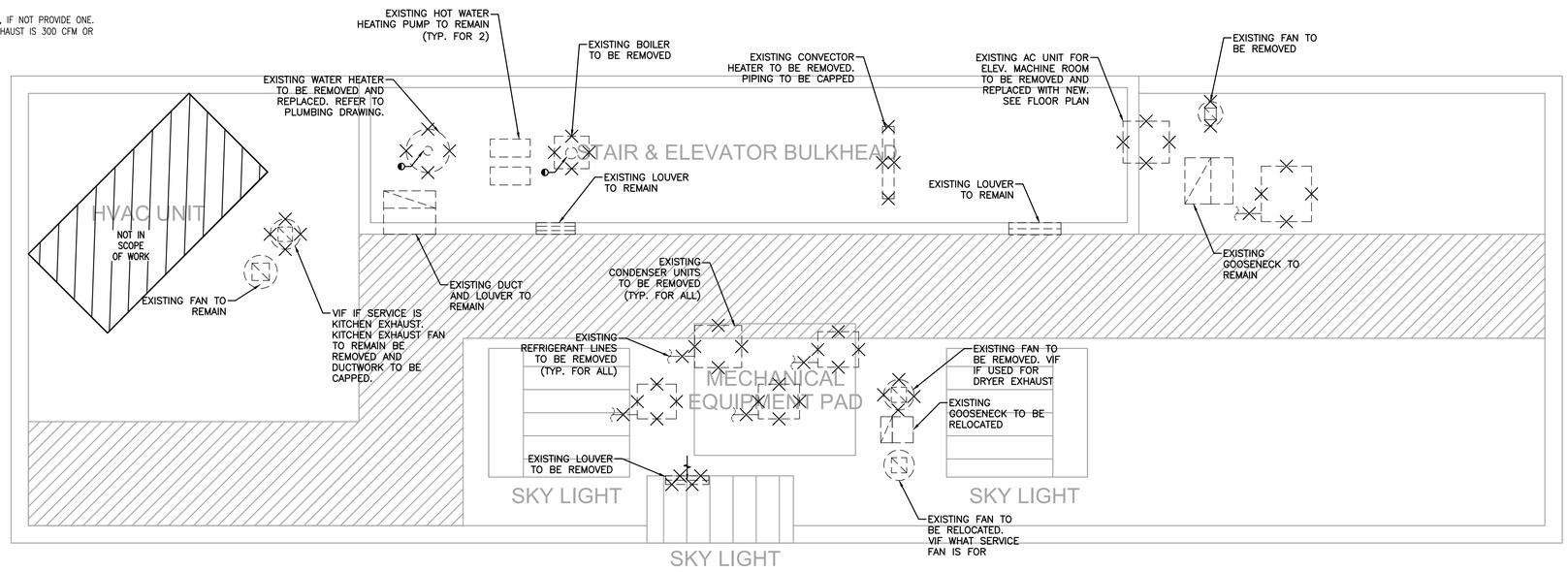
N.Y.C D.O.B No. :
M01244603-S1

DUCTWORK NOTES:

1. PROVIDE 18 GAGE ALUMINUM DUCT FOR CLOTH DRYER EXHAUST
2. PROVIDE 18 GA. DUCTWORK FOR RESIDENTIAL KITCHEN EXHAUST
3. PROVIDE MIN. 24 GAGE FOR GENERAL DUCTWORK
4. ALL INTAKE OPENING LOCATIONS SHALL COMPLY WITH 2022 MC SECTION 401.4
5. ALL EXHAUST OUTLET LOCATION SHALL COMPLY WITH 2022 MC SECTION 501.3.1
6. PROVIDE MIN. 1.5" INSULATION FOR ALL DUCTWORK DEDICATED FOR OA
7. PROVIDE MIN. 1.5" INSULATION FOR ALL CONCEALED DUCTWORK DEDICATED FOR AC SUPPLY
8. PROVIDE 1.5" ACOUSTICAL LINING FOR AC DUCTWORK
9. ALL INTERIOR AND EXTERIOR MECHANICAL EQUIPMENT SHALL COMPLY WITH 2022 MC SECTION 928
10. ALL DUCTWORK WITH A CROSS-SECTIONAL AREA OF 20 IN^2 OR LESS PASSING THROUGH FIRE-RATED ASSEMBLIES EXCEPT FIRE WALLS SHALL NOT REQUIRE FIRE DAMPERS OR FIRE SMOKE DAMPERS, AS PER 2022 MC 607.1.1 & 2022 BC SECTION 716.5
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HVAC PIPING NOTES:

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2. CONDENSATE PIPE MATERIAL AND SIZING SHALL COMPLY WITH 2022 MC SECTION 307.2.2.
3. PROVIDE THERMAL EXPANSION LOOP FOR REFRIGERANT PIPING AT EVERY 100FT OF RUN.
4. PROVIDE CONDENSATE PUMP FOR ALL AC UNITS (WITH THE EXCEPTION OF WALL MOUNTED UNITS AND HPAC UNITS) AND ROUTE CONDENSATE TO NEAREST INDIRECT DRAIN
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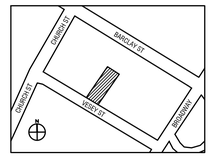


ROOF PLAN
SCALE: 1/4"=1'-0"

ACE
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EMAIL: LUCAS@SIHNYC.ORG

SITE PLAN



DoB STAMPS:

No.	DATE	REVISION
	01/08/26	BID SET

NOTE: Drawing may be printed at reduced scale

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PROJECT:

18 VESEY STREET
MANHATTAN, NY
BLOCK: 88 LOT: 4

DRAWING TITLE:

EXISTING ROOF

DATE:	01/06/2026
PROJECT No.:	25086
CHECKED BY:	
DRAWN BY:	PY/LZ/BL

SEAL AND SIGNATURE

DWG. No.:
M053.00

Sheets: 04 of 18

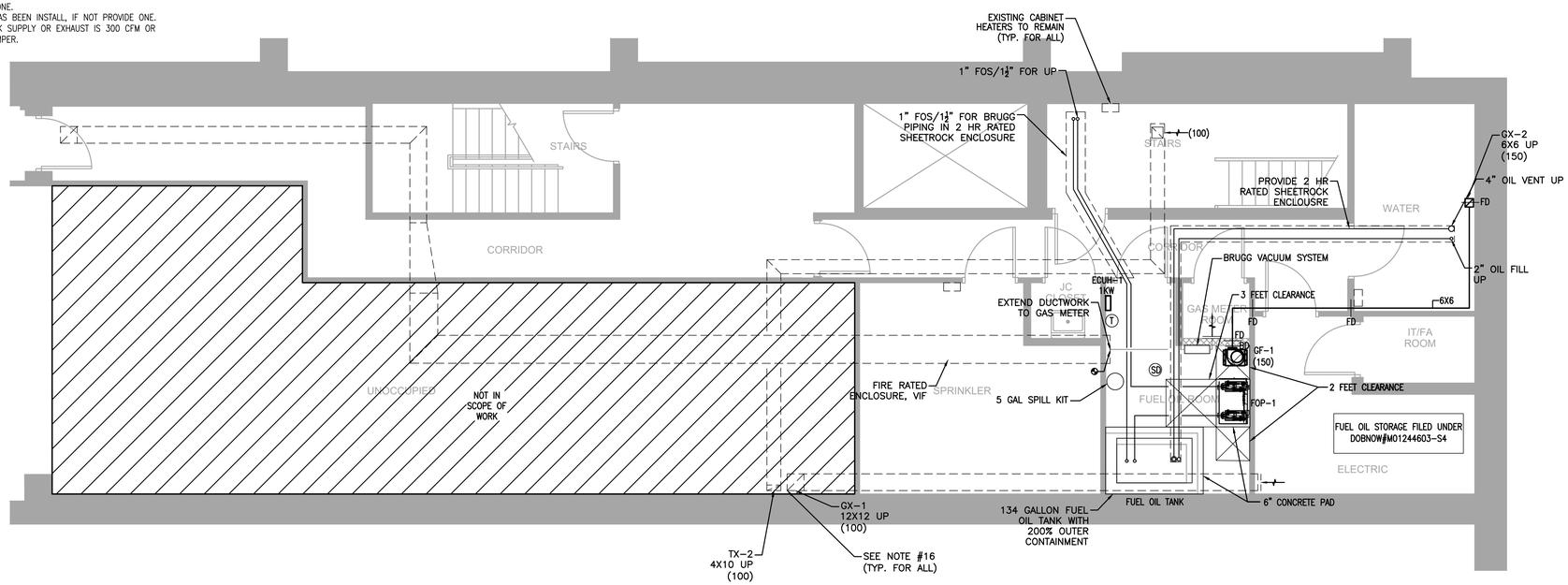
N.Y.C D.O.B No. :
M01244603-S1

DUCTWORK NOTES:

1. PROVIDE 18 GAGE ALUMINUM DUCT FOR CLOTH DRYER EXHAUST
2. PROVIDE 18 GA. DUCTWORK FOR RESIDENTIAL KITCHEN EXHAUST
3. PROVIDE MIN. 24 GAGE FOR GENERAL DUCTWORK
4. ALL INTAKE OPENING LOCATIONS SHALL COMPLY WITH 2022 MC SECTION 401.4
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10. ALL DUCTWORK WITH A CROSS-SECTIONAL AREA OF 20 IN^2 OR LESS PASSING THROUGH FIRE-RATED ASSEMBLIES EXCEPT FIRE WALLS SHALL NOT REQUIRE FIRE DAMPERS OR FIRE SMOKE DAMPERS, AS PER 2022 MC 607.1.1 & 2022 BC SECTION 716.5
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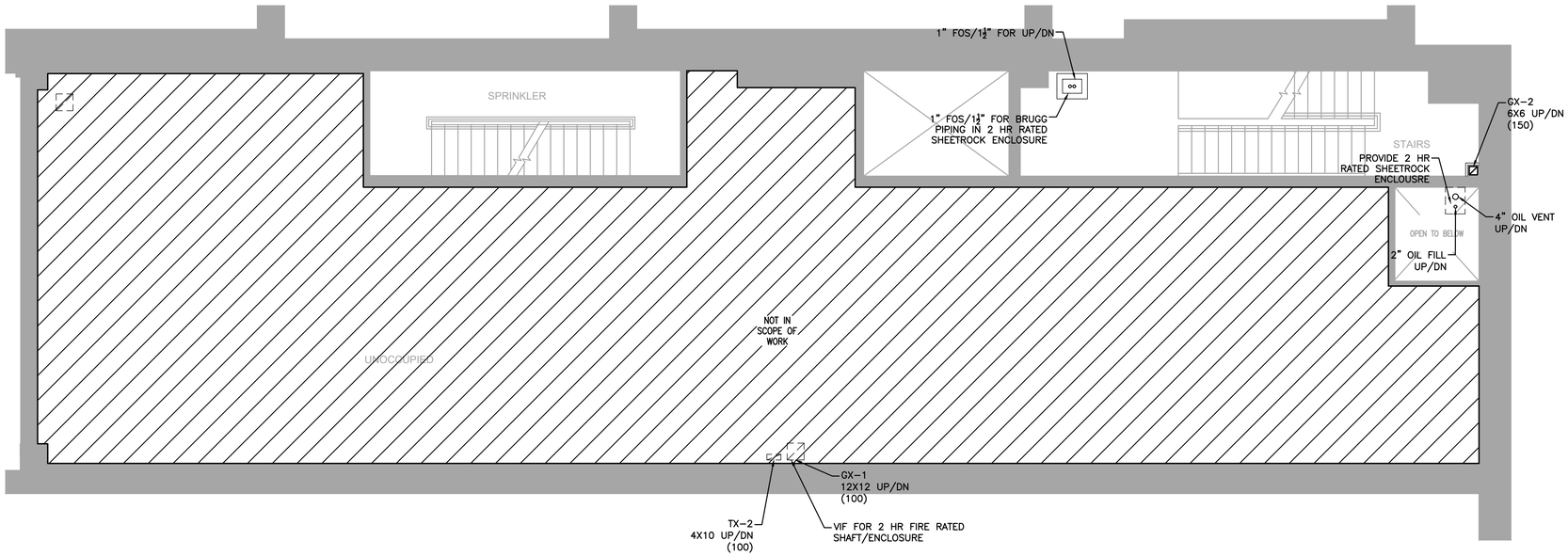
HVAC PIPING NOTES:

1. PROVIDE PROPER INSULATION THICKENESS FOR REFRIGERANT PIPING AS PER NYCECC TABLE C403.11.3 AND/OR 2016 ASHRAE TABLE 6.8.3.1
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SUB-CELLAR FLOOR PLAN

SCALE: 1/4"=1'-0"



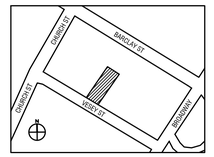
CELLAR FLOOR PLAN

SCALE: 1/4"=1'-0"

ALL CITY ENGINEERING, P.C.
33 Great Neck Road #3
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(516)321-9490
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NEW YORK, NY 1003
TEL: (212) 677-4800
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SITE PLAN



DoB STAMPS:

No.	DATE	REVISION
	01/08/26	BID SET

NOTE: Drawing may be printed at reduced scale

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PROJECT:

18 VESEY STREET
MANHATTAN, NY
BLOCK: 88 LOT: 4

DRAWING TITLE:

PROPOSED SUB-CELLAR AND CELLAR FLOOR PLAN

DATE:	01/06/2026
PROJECT No.:	25086
CHECKED BY:	
DRAWN BY:	PY/LZ/BL

SEAL AND SIGNATURE

DWG. No.:
M101.00

Sheets: 05 of 18

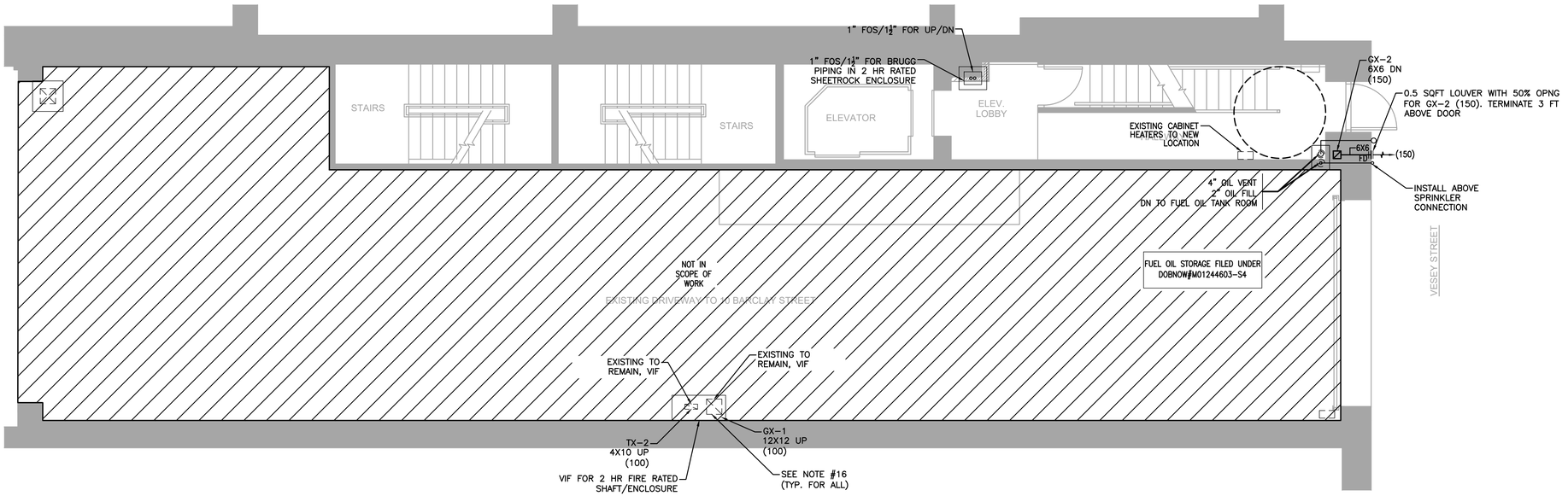
N.Y.C D.O.B No. :
M01244603-S1

DUCTWORK NOTES:

1. PROVIDE 18 GAGE ALUMINUM DUCT FOR CLOTH DRYER EXHAUST
2. PROVIDE 18 GA. DUCTWORK FOR RESIDENTIAL KITCHEN EXHAUST
3. PROVIDE MIN. 24 GAGE FOR GENERAL DUCTWORK
4. ALL INTAKE OPENING LOCATIONS SHALL COMPLY WITH 2022 MC SECTION 401.4
5. ALL EXHAUST OUTLET LOCATION SHALL COMPLY WITH 2022 MC SECTION 501.3.1
6. PROVIDE MIN. 1.5" INSULATION FOR ALL DUCTWORK DEDICATED FOR OA
7. PROVIDE MIN. 1.5" INSULATION FOR ALL CONCEALED DUCTWORK DEDICATED FOR AC SUPPLY
8. PROVIDE 1.5" ACOUSTICAL LINING FOR AC DUCTWORK
9. ALL INTERIOR AND EXTERIOR MECHANICAL EQUIPMENT SHALL COMPLY WITH 2022 MC SECTION 928
10. ALL DUCTWORK WITH A CROSS-SECTIONAL AREA OF 20 IN^2 OR LESS PASSING THROUGH FIRE-RATED ASSEMBLIES EXCEPT FIRE WALLS SHALL NOT REQUIRE FIRE DAMPERS OR FIRE SMOKE DAMPERS, AS PER 2022 MC 607.1.1 & 2022 BC SECTION 716.5
11. PROVIDE ACCESS DOOR FOR ALL FIRE DAMPER, FIRE/SMOKE DAMPER AND POST-FIRE SMOKE DAMPER
12. ALL OUTSIDE AIR INTAKE AND EXHAUST MOTORIZED DAMPERS SHALL BE CLASS I LEAKAGE TYPE
13. DUCT CONSTRUCTION AND INSTALLATION SHALL STRICTLY FOLLOW ANSI/SMACNA 006-2020 STANDARD.
14. PROVIDE VOLUME DAMPER/CORD OPERATED DAMPER FOR EVERY BRANCH OF DUCTWORK.
15. VERIFY IN FIELD IF EXISTING SUPPLY DUCTS HAVE VOLUME DAMPERS, IF NOT PROVIDE ONE.
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HVAC PIPING NOTES:

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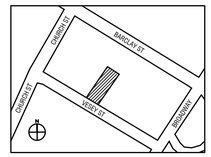


FIRST FLOOR PLAN
SCALE: 1/4"=1'-0"

ACE
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EMAIL: LUCAS@SIHNYC.ORG

SITE PLAN



DoB STAMPS:

No.	DATE	REVISION
	01/08/26	BID SET

NOTE: Drawing may be printed at reduced scale

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PROJECT:

18 VESEY STREET
MANHATTAN, NY
BLOCK: 88 LOT: 4

DRAWING TITLE:

PROPOSED FIRST FLOOR PLAN

DATE:	01/06/2026
PROJECT No.:	25086
CHECKED BY:	
DRAWN BY:	PY/LZ/BL

SEAL AND SIGNATURE

DWG. No.:
M102.00

Sheets: 06 of 18

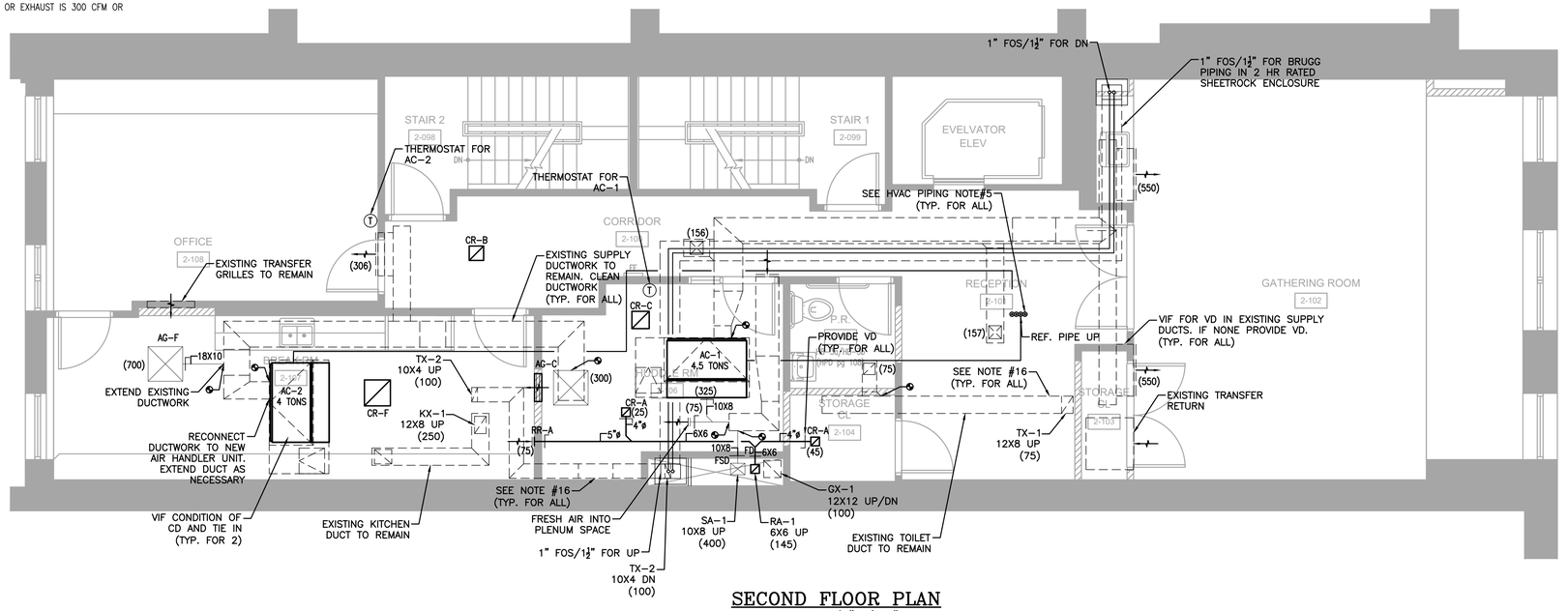
N.Y.C D.O.B No. :
M01244603-S1

DUCTWORK NOTES:

1. PROVIDE 18 GAGE ALUMINUM DUCT FOR CLOTH DRYER EXHAUST
2. PROVIDE 18 GA. DUCTWORK FOR RESIDENTIAL KITCHEN EXHAUST
3. PROVIDE MIN. 24 GAGE FOR GENERAL DUCTWORK
4. ALL INTAKE OPENING LOCATIONS SHALL COMPLY WITH 2022 MC SECTION 401.4
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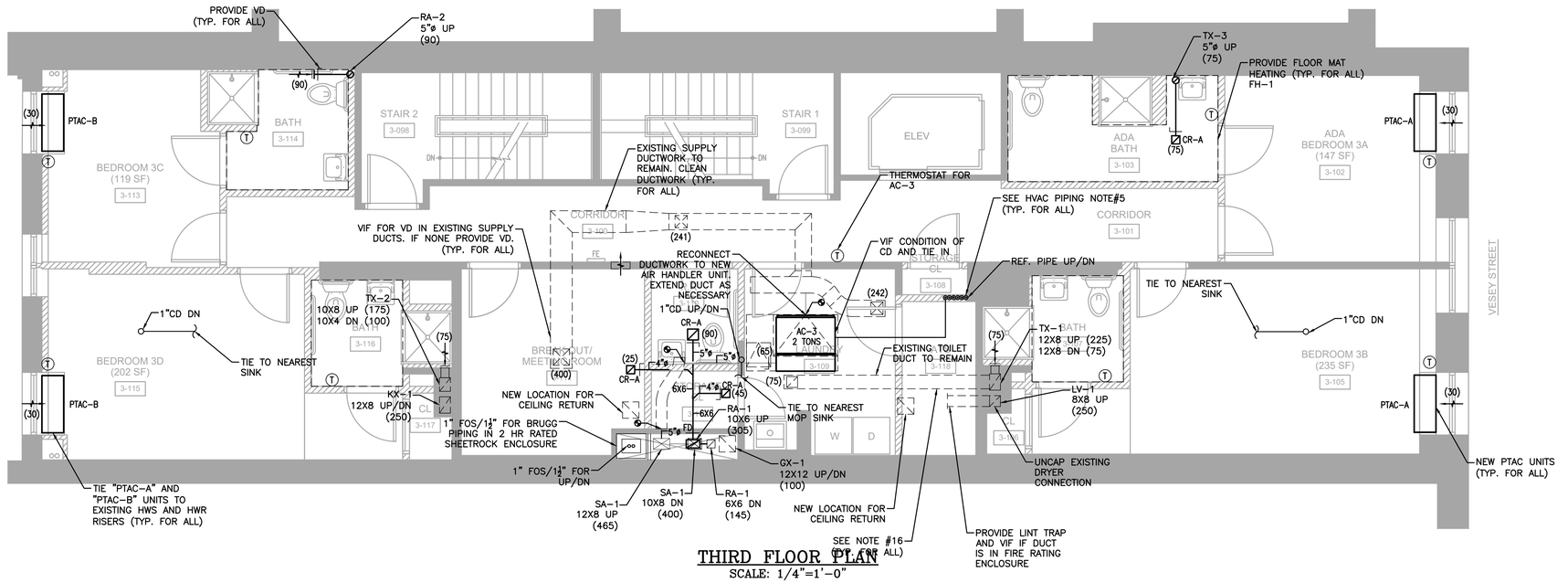
HVAC PIPING NOTES:

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3. PROVIDE THERMAL EXPANSION LOOP FOR REFRIGERANT PIPING AT EVERY 100FT OF RUN.
4. PROVIDE CONDENSATE PUMP FOR ALL AC UNITS (WITH THE EXCEPTION OF WALL MOUNTED UNITS AND HPAC UNITS) AND ROUTE CONDENSATE TO NEAREST INDIRECT DRAIN
5. PER ADDENDUM TO ASHRAE STANDARD 15-2024. VENTILATION FOR SHAFTS CONTAINING REFRIGERANT PIPING USING R-32 AND R-454B REFRIGERANT SHALL NOT BE REQUIRED WHERE ALL THE REFRIGERANT PIPE OR TUBE IS CONTINUOUS AND HAS BEEN TESTED IN ACCORDANCE WITH SECTION 9.13 OF THE SAFETY STANDARD FOR REFRIGERATION SYSTEMS AS WELL AS BEING TESTED AT 650 PSI OPERATING PRESSURE FOR 24 HOURS.



SECOND FLOOR PLAN

SCALE: 1/4"=1'-0"



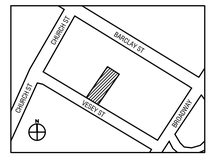
THIRD FLOOR PLAN

SCALE: 1/4"=1'-0"

ALL CITY ENGINEERING, P.C.
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(516)321-9490
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NEW YORK, NY 1003
TEL: (212) 677-4800
EMAIL: LUCAS@SIHNYC.ORG

SITE PLAN



DoB STAMPS:

No.	DATE	REVISION
	01/08/26	BID SET

NOTE: Drawing may be printed at reduced scale

IT IS A VIOLATION OF THE STATE EDUCATION LAW SECTION 7209 (2) FOR ANY PERSON TO ALTER AN ITEM IN ANY WAY UNLESS SUCH PERSON IS ACTING UNDER THE DIRECTION OF A LICENSED PROFESSIONAL ENGINEER, AND THE ENGINEER STAMPS SUCH CHANGES

PROJECT:

18 VESEY STREET
MANHATTAN, NY
BLOCK: 88 LOT: 4

DRAWING TITLE:

PROPOSED SECOND AND THIRD FLOOR PLAN

DATE:	01/06/2026
PROJECT No.:	25086
CHECKED BY:	
DRAWN BY:	PY/LZ/BL

SEAL AND SIGNATURE

DWG. No.:
M103.00

Sheets: 07 of 18

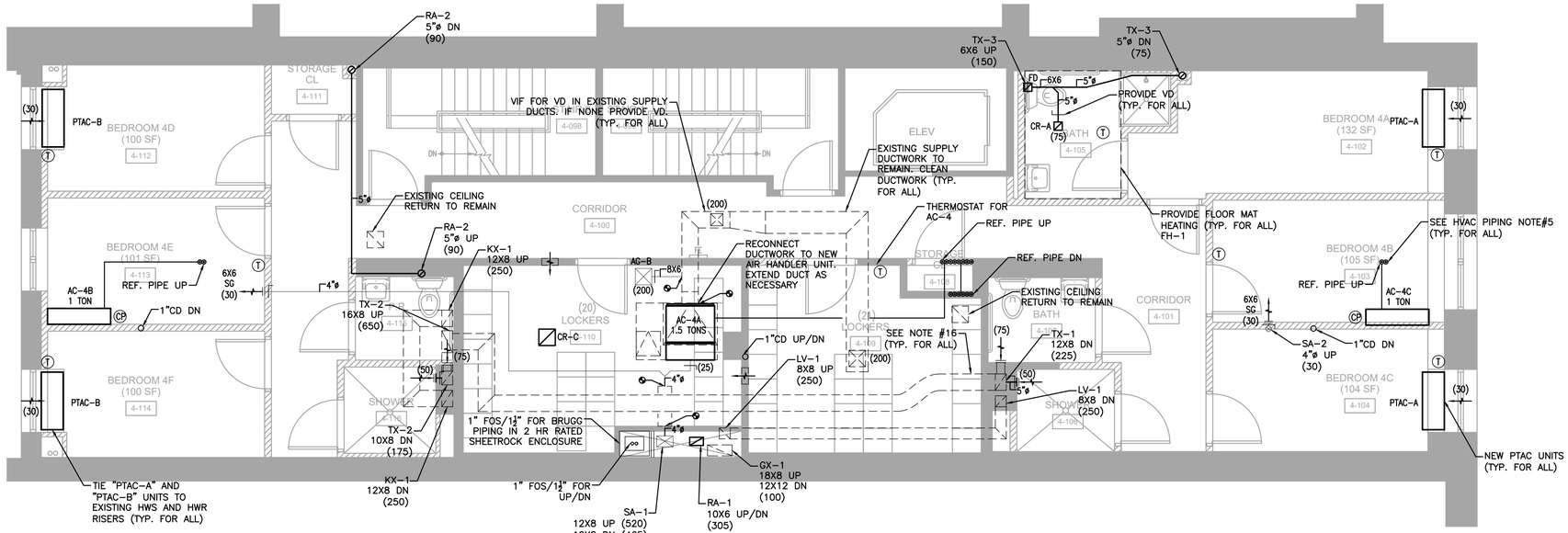
N.Y.C D.O.B No. :
M01244603-S1

DUCTWORK NOTES:

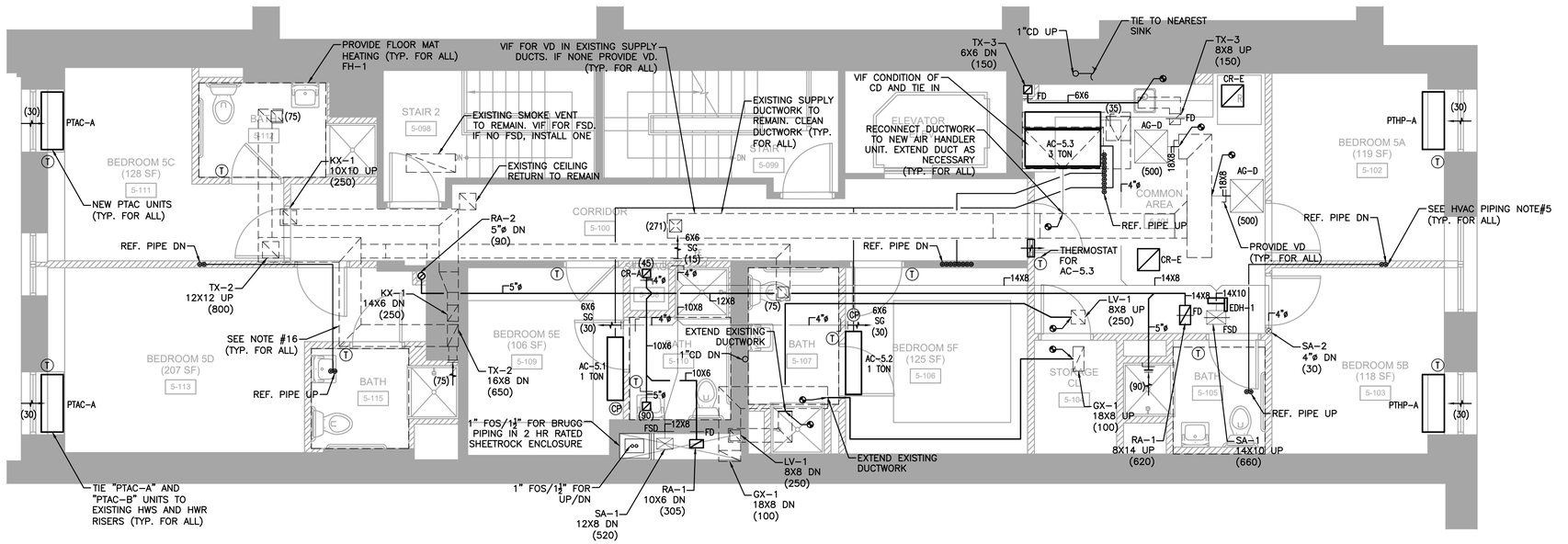
1. PROVIDE 18 GAGE ALUMINUM DUCT FOR CLOTH DRYER EXHAUST
2. PROVIDE 18 GA. DUCTWORK FOR RESIDENTIAL KITCHEN EXHAUST
3. PROVIDE MIN. 24 GAGE FOR GENERAL DUCTWORK
4. ALL INTAKE OPENING LOCATIONS SHALL COMPLY WITH 2022 MC SECTION 401.4
5. ALL EXHAUST OUTLET LOCATION SHALL COMPLY WITH 2022 MC SECTION 501.3.1
6. PROVIDE MIN. 1.5" INSULATION FOR ALL DUCTWORK DEDICATED FOR OA
7. PROVIDE MIN. 1.5" INSULATION FOR ALL CONCEALED DUCTWORK DEDICATED FOR AC SUPPLY
8. PROVIDE 1.5" ACOUSTICAL LINING FOR AC DUCTWORK
9. ALL INTERIOR AND EXTERIOR MECHANICAL EQUIPMENT SHALL COMPLY WITH 2022 MC SECTION 928
10. ALL DUCTWORK WITH A CROSS-SECTIONAL AREA OF 20 IN^2 OR LESS PASSING THROUGH FIRE-RATED ASSEMBLIES EXCEPT FIRE WALLS SHALL NOT REQUIRE FIRE DAMPERS OR FIRE SMOKE DAMPERS, AS PER 2022 MC 607.1.1 & 2022 BC SECTION 716.5
11. PROVIDE ACCESS DOOR FOR ALL FIRE DAMPER, FIRE/SMOKE DAMPER AND POST-FIRE SMOKE DAMPER
12. ALL OUTSIDE AIR INTAKE AND EXHAUST MOTORIZED DAMPERS SHALL BE CLASS I LEAKAGE TYPE
13. DUCT CONSTRUCTION AND INSTALLATION SHALL STRICTLY FOLLOW ANSI/SMACNA 006-2020 STANDARD.
14. PROVIDE VOLUME DAMPER/CORD OPERATED DAMPER FOR EVERY BRANCH OF DUCTWORK.
15. VERIFY IN FIELD IF EXISTING SUPPLY DUCTS HAVE VOLUME DAMPERS, IF NOT PROVIDE ONE.
16. VERIFY IN FIELD IF EXISTING DUCTWORK HAS FIRE DAMPER AND FIRE SMOKE DAMPER HAS BEEN INSTALL, IF NOT PROVIDE ONE.
17. VERIFY IN FIELD IF MOTORIZED/BACKDRAFT DAMPERS ARE INSTALLED. IF FAN/GOOSENECK SUPPLY OR EXHAUST IS 300 CFM OR MORE PROVIDE MOTORIZED DAMPER AND IF 300 CFM OR LESS PROVIDE BACKDRAFT DAMPER.

HVAC PIPING NOTES:

1. PROVIDE PROPER INSULATION THICKENESS FOR REFRIGERANT PIPING AS PER NYCECC TABLE C403.11.3 AND/OR 2016 ASHRAE TABLE 6.8.3.1
2. CONDENSATE PIPE MATERIAL AND SIZING SHALL COMPLY WITH 2022 MC SECTION 307.2.2.
3. PROVIDE THERMAL EXPANSION LOOP FOR REFRIGERANT PIPING AT EVERY 100FT OF RUN.
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FOURTH FLOOR PLAN
SCALE: 1/4"=1'-0"

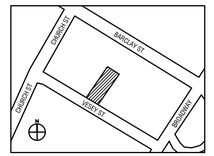


FIFTH FLOOR PLAN
SCALE: 1/4"=1'-0"

ACE
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SITE PLAN



DoB STAMPS:

No.	DATE	REVISION
	01/08/26	BID SET

NOTE: Drawing may be printed at reduced scale

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PROJECT:

18 VESEY STREET
MANHATTAN, NY
BLOCK: 88 LOT: 4

DRAWING TITLE:

PROPOSED FOURTH AND FIFTH FLOOR PLAN

DATE:	01/06/2026
PROJECT No.:	25086
CHECKED BY:	
DRAWN BY:	PY/LZ/BL

SEAL AND SIGNATURE

DWG. No.:
M104.00

Sheets: 08 of 18

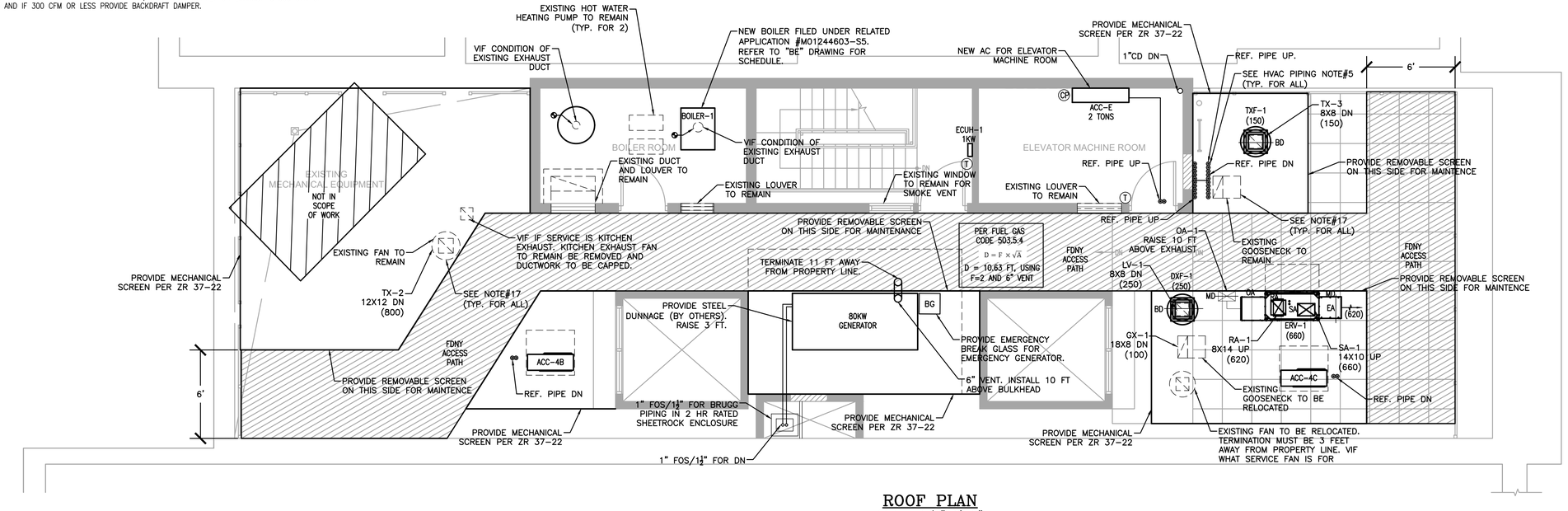
N.Y.C D.O.B No. :
M01244603-S1

DUCTWORK NOTES:

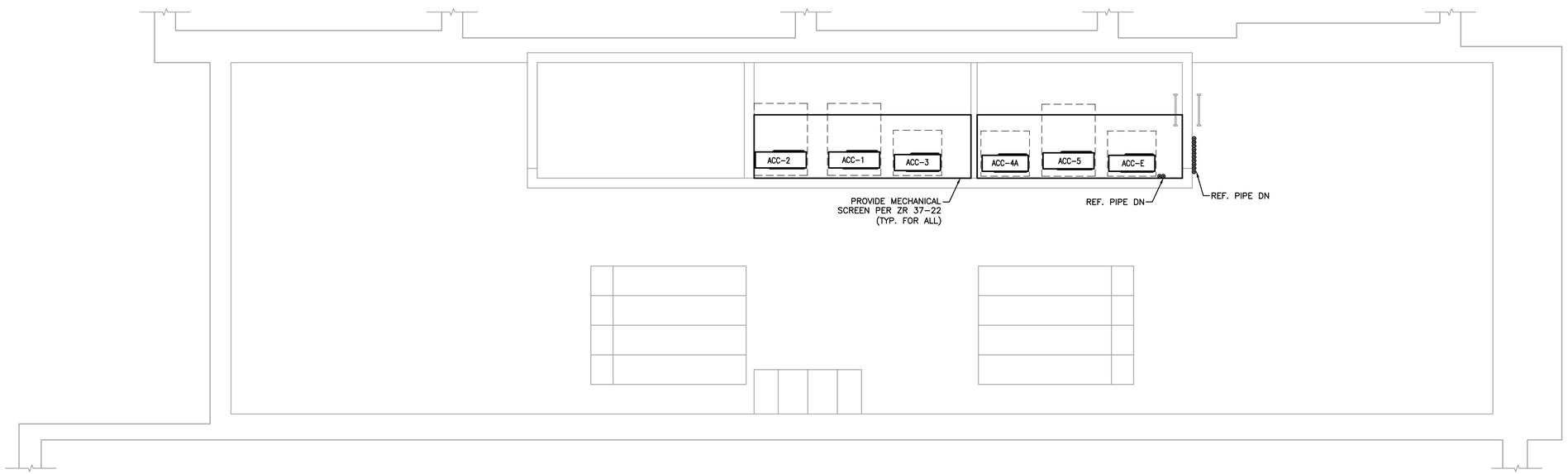
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ROOF PLAN
SCALE: 1/4"=1'-0"

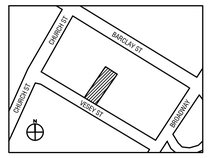


BULKHEAD PLAN
SCALE: 1/4"=1'-0"

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SITE PLAN



DoB STAMPS:

No.	DATE	REVISION
	01/08/26	BID SET

NOTE: Drawing may be printed at reduced scale

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PROJECT:

18 VESEY STREET
MANHATTAN, NY
BLOCK: 88 LOT: 4

DRAWING TITLE:

PROPOSED ROOF AND BULKHEAD PLAN

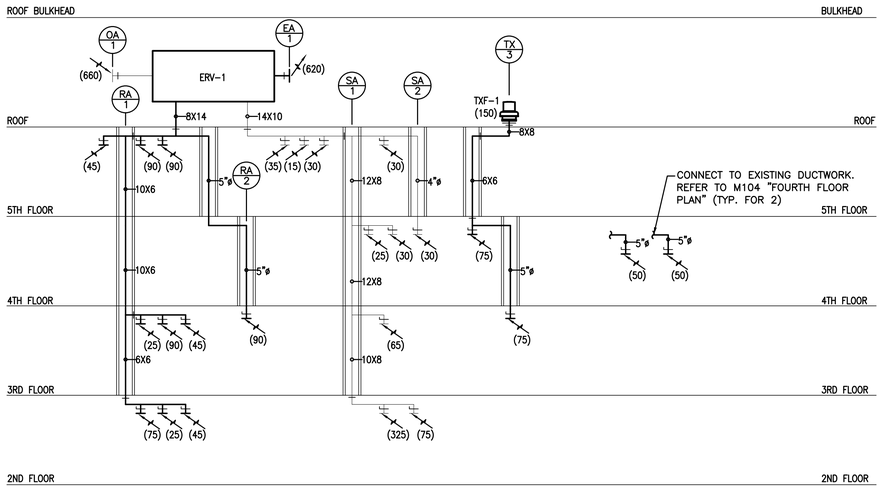
DATE:	01/06/2026
PROJECT No.:	25086
CHECKED BY:	
DRAWN BY:	PY/LZ/BL

SEAL AND SIGNATURE

DWG. No.:
M105.00

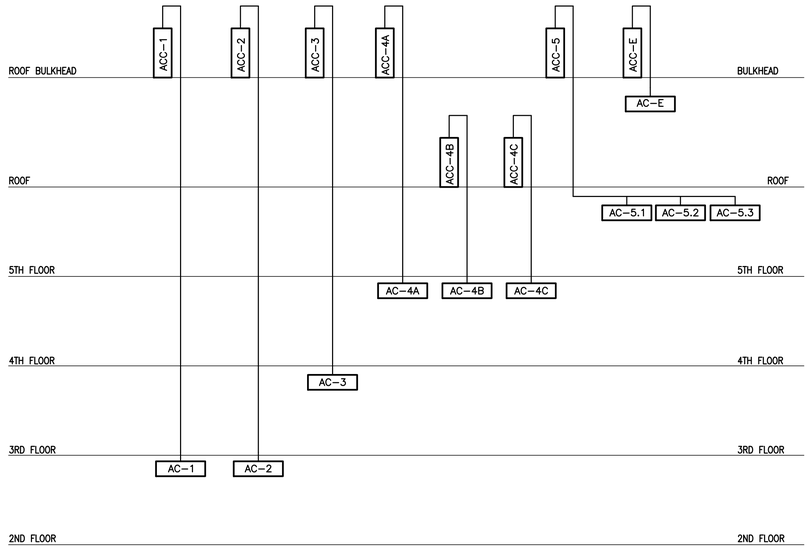
Sheets: 09 of 18

N.Y.C D.O.B No. :
M01244603-S1



MECHANICAL RISER DIAGRAM – AIR

SCALE: NTS



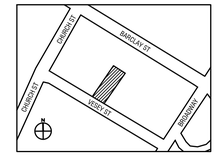
MECHANICAL RISER DIAGRAM – REFRIGERANT PIPE

SCALE: NTS

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SITE PLAN



DoB STAMPS:

No.	DATE	REVISION
	01/08/26	BID SET

NOTE: Drawing may be printed at reduced scale

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PROJECT:

18 VESEY STREET
MANHATTAN, NY
BLOCK: 88 LOT: 4

DRAWING TITLE:

MECHANICAL AIR AND REFRIGERANT PIPE RISER DIAGRAM

DATE:	01/06/2026
PROJECT No.:	25086
CHECKED BY:	
DRAWN BY:	PY/LZ/BL

SEAL AND SIGNATURE

DWG. No.:
M301.00

Sheets: 10 of 18

N.Y.C D.O.B No. :
M01244603-S1

SINGLE-SPLIT SYSTEM HEAT PUMP CONDENSING UNIT SCHEDULE

"APPROVAL TEST AGENCY" TEST IS BASED ON AHRI 210/240

CONDENSING UNIT																							
UNIT TAG	COOLING CAPCACITY (BTU/HR)	HEATING CAPACITY (BTU/HR)	REF. TYPE	EFFICIENCY		ELECTRICAL DATA					REFRIGERANT LINES (IN.)		WEIGHT (LBS)	MANUFACTURER	MODEL NUMBER	SERIAL NUMBER	UNIT TAG	LOCATION	REF. TYPE	ELECTRICAL DATA					MODEL NUMBER
				SEER	HSPF	VOLTS	PHASE	HZ	MCA	MOP	LIQUID	GAS								VOLTS	PHASE	HZ	MCA	MOP	
ACC-1	60,000	66,000	R-454B	16.40	9.50	208	1	60	45	80	3/8	3/4	278	MITSUBISHI	MXZ-SM60NL	–	AC-1	SECOND FLOOR	R-454B	208	1	60	3.38	15	PEFY-L54NMAU-A
ACC-2	48,000	54,000	R-454B	18	10	208	1	60	45	80	3/8	5/8	283	MITSUBISHI	MXZ-SM48NLHZ	–	AC-2	SECOND FLOOR	R-454B	208	1	60	3.38	15	PEFY-L48NMAU-A
ACC-3	21,200	28,000	R-454B	19.6	9.1	208	1	60	22	37	3/8	5/8	155	MITSUBISHI	PUZ-AH24NL	–	AC-3	THIRD FLOOR	R-454B	208	1	60	2.25	15	PEAD-AA24NL
ACC-4A	18,000	22,000	R-454B	20.1	8.8	208	1	60	16	27	1/4	1/2	60	MITSUBISHI	PUZ-AK18NL	–	AC-4A	FOURTH FLOOR	R-454B	208	1	60	2.25	15	PEAD-AA18NL
ACC-4B	12,000	14,400	R-454B	25.6	10.7	208	1	60	12	15	1/4	3/8	77	MITSUBISHI	MUZ-GX12NL	–	AC-4B	FOURTH FLOOR	R-454B	208	1	60	1	15	MSZ-GX12NL
ACC-4C	12,000	14,400	R-454B	25.6	10.7	208	1	60	12	15	1/4	3/8	77	MITSUBISHI	MUZ-GX12NL	–	AC-4C	FOURTH FLOOR	R-454B	208	1	60	1	15	MSZ-GX12NL
ACC-E	22,400	27,600	R-454B	21.5	10.3	208	1	60	23	25	1/4	5/8	37	MITSUBISHI	MUZ-GX24NL	–	AC-E	EMR	R-454B	208	1	60	1	15	MSZ-GX24NL

NOTE: PROVIDE LOW AMBIENT KIT

MULTI-SPLIT SYSTEM HEAT PUMP CONDENSING UNIT SCHEDULE

"APPROVAL TEST AGENCY" TEST IS BASED ON AHRI 210/240

UNIT TAG	COOLING CAPCACITY (BTU/HR)	HEATING CAPACITY (BTU/HR)	REF. TYPE	EFFICIENCY		RELATED AIR HANDLER	ELECTRICAL DATA					REFRIGERANT LINES (IN.)		WEIGHT (LBS)	MANUFACTURER	MODEL NUMBER	SERIAL NUMBER
				SEER	HSPF		VOLTS	PHASE	HZ	MCA	MOP	LIQUID	GAS				
ACC-5	60,000	66,000	R-454B	16.40	9.50	AC-5.1~5.3	208	1	60	45	80	3/8	3/4	278	MITSUBISHI	MXZ-SM60NL	–

MULTI-SPLIT SYSTEM INDOOR UNIT SCHEDULE

"APPROVAL TEST AGENCY" TEST IS BASED ON AHRI 210/240

UNIT TAG	LOCATION	COOLING CAPCACITY (BTU/HR)	HEATING CAPACITY (BTU/HR)	REF. TYPE	ELECTRICAL DATA					DRAIN CONNECTION (IN.)	MANUFACTURER	MODEL NUMBER
					VOLTS	PHASE	HZ	MCA	MOP			
AC-5.1	FIFTH FLOOR	12,000	13,500	R-454B	208	1	60	0.24	15	1-1/4	MITSUBISHI	PKFY-L12NLMU-A
AC-5.2	FIFTH FLOOR	12,000	13,500	R-454B	208	1	60	0.24	15	1-1/4	MITSUBISHI	PKFY-L12NLMU-A
AC-5.3	FIFTH FLOOR	36,000	40,000	R-454B	208	1	60	3.13	15	1-1/4	MITSUBISHI	PEFY-L36NMAU-A

PACKAGED TERMINAL AIR CONDITIONING UNIT SCHEDULE

UNIT TAG	LOCATION	REF.	NOMINAL CAPACITY		ELECTRIC HEAT (KW)	FRESH AIR VENT (CFM)	EFFICIENCY	ELECTRICAL DATA			UNIT DIMENSIONS			SOUND PRESSURE dB(A)	WEIGHT (LBS)	MANUFACTURER	MODEL
			TONNAGE	COOLING (MBH)			EER	V/ø/HZ	MCA	MOCP	HEIGHT (IN.)	WIDTH (IN.)	DEPTH (IN.)				
PTAC-A	SEE PLANS	R32	1	12.7	1.50	**60**	10.60	115/1/60	9.9	20.0	**16-3/4**	42	19-1/4	–	124	ICE AIR	5RSCC13
PTAC-B	SEE PLANS	R32	0.75	9.0	1.50	**60**	11.40	115/1/60	9.6	20.0	**16-3/4**	42	20-1/4	–	134	ICE AIR	5RSNU09

NOTES:
1. UNIT SHALL COME WITH FACTORY INSTALLED MANUAL OUTSIDE AIR VENT – MECHANICAL CONTRACTOR TO VERIFY THAT VENT IS OPERATIONAL PRIOR TO INSTALLATION
2. PROVIDE WALL MOUNTED 7 DAY PROGRAMMABLE THERMOSTAT
3. PROVIDE WASHABLE FILTER
4. PROVIDE INTERNAL DRAIN SYSTEM FOR CONDENSATE
5. COORDINATE EXTERIOR GRILLE FINISH WITH ARCHITECT
6. UNITS SHALL BE CERTIFIED TO AHRI STANDARD 310/380

PACKAGED TERMINAL HEAT PUMP UNIT SCHEDULE

UNIT TAG	LOCATION	REF.	NOMINAL CAPACITY			BACKUP ELECTRIC HEAT (KW)	FRESH AIR VENT (CFM)	EFFICIENCY		ELECTRICAL DATA			UNIT DIMENSIONS			SOUND PRESSURE dB(A)	WEIGHT (LBS)	MANUFACTURER	MODEL
			TONNAGE	COOLING (MBH)	HEATING (MBH)			EER	COP (47°F)	V/ø/HZ	MCA	MOCP	HEIGHT (IN.)	WIDTH (IN.)	DEPTH (IN.)				
PTHP-A	SEE PLANS	R32	1	12.5	13.7	4.30	**60**	11.10	3.70	208/1/60	26.4	30.0	**16**	42	21-1/8	–	134	ICE AIR	RSXC13

NOTES:
1. UNIT SHALL COME WITH FACTORY INSTALLED MANUAL OUTSIDE AIR VENT – MECHANICAL CONTRACTOR TO VERIFY THAT VENT IS OPERATIONAL PRIOR TO INSTALLATION
2. PROVIDE WALL MOUNTED 7 DAY PROGRAMMABLE THERMOSTAT
3. PROVIDE WASHABLE FILTER
4. PROVIDE INTERNAL DRAIN SYSTEM FOR CONDENSATE
5. COORDINATE EXTERIOR GRILLE FINISH WITH ARCHITECT
6. UNITS SHALL BE CERTIFIED TO AHRI STANDARD 310/380

FAN SCHEDULE

UL LISTED 705

UNIT TAG	SERVICE	LOCATION	FAN DATA				MOTOR DATA				MANUFACTURER	MODEL NUMBER	NOTES
			CAPACITY (CFM)	STATIC PRESSURE (IN. WC)	DRIVE TYPE	FEI	RATED HP	VOLTS	PHASE	HZ			
GF-1	FUEL OIL STORAGE RM	SUB-CELLAR	150	1.00	DIRECT	–	172W	120	1	60	GREENHECK	SP-B200	
DXF-1	DRYER	ROOF	250	1.00	DIRECT	–	0.25	120	1	60	GREENHECK	CUE-100HP-VG	
TXF-1	BATHROOM	ROOF	150	1.00	BELT	–	1/6	120	1	60	GREENHECK	GB-097-6	

NOTES:
1. PROVIDE LOW LEAKAGE DAMPER FOR ALL EXHAUST FANS.
2. PROVIDE BACKDRAFT DAMPER FOR ALL FANS SUPPLYING OR EXHAUSTING LESS THAN 300 CFM
3. PROVIDE MOTORIZED DAMPER FOR ALL FANS SUPPLYING OR EXHAUSTING 300 CFM OR MORE
4. PROVIDE HIGH EFFICIENCY MOTORS

FLOOR HEAT MAT SCHEDULE

UNIT TAG	LOCATION	ELECTRICAL DATA			MANUFACTURER	MODEL NUMBER
		V/ø/Hz	W/SF	AMP/SF		
FH-1	SEE PLANS	120/1/60	12.00	0.10	KING	FCM-1

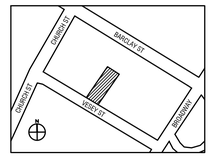
ELECTRIC DUCT HEATER SCHEDULE

UNIT TAG	SIZE (IN.)	HEATING CAPACITY		AIR DATA		ELECTRICAL DATA				MANUFACTURER	MODEL	TYPE
		BTU/HR	KW	CAPACITY (CFM)	TEMPERATURE CHANGE (°F)	VOLTS	PHASE	HZ	AMPS			
EDH-1	14X10	27296	8	660	38	**208**	**1**	**60**	**38.5**	INDEECO	QUA	SLIP-IN

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OWNER:
SEAFARERS INTERNATIONAL HOUSE
123 EAST 15TH STREET
NEW YORK, NY 1003
TEL: (212) 677-4800
EMAIL: LUCAS@SIHNYC.ORG

SITE PLAN



DoB STAMPS:

No.	DATE	REVISION
	01/08/26	BID SET

NOTE: Drawing may be printed at reduced scale

IT IS A VIOLATION OF THE STATE EDUCATION LAW SECTION 7209 (2) FOR ANY PERSON TO ALTER AN ITEM IN ANY WAY UNLESS SUCH PERSON IS ACTING UNDER THE DIRECTION OF A LICENSED PROFESSIONAL ENGINEER, AND THE ENGINEER STAMPS SUCH CHANGES

PROJECT:

18 VESEY STREET
MANHATTAN, NY
BLOCK: 88 LOT: 4

DRAWING TITLE:

MECHANICAL EQUIPMENT SCHEDULE

DATE: 01/06/2026
PROJECT No.: 25086
CHECKED BY:
DRAWN BY: PY/LZ/BL
SEAL AND SIGNATURE

DWG. No.:
M401.00

Sheets: 11 of 18

N.Y.C D.O.B No. :
M01244603-S1

SUPPLY AIR GRILLES SCHEDULE (CEILING MOUNTED)

TAG	NOMINAL DUCT SIZE (IN)	GRILL SIZE (IN.)	CFM RANGE	MAXIMUM DUCT VELOCITY FPM	PRESSURE LOSS (IN. WG)	THROW (FT)	NC LEVEL	MODEL	MANUFACTURER
CG-A	6X6	12X12	0-125	<500	0.117	6-7-10	<20	TDC	TITUS
CG-B	9X9	12X12	126-250	<500	0.117	8-11-16	<20	TDC	TITUS
CG-C	12X12	24X24	251-400	<400	0.0750	9-13-19	<20	TDC	TITUS
CG-D	15X15	24X24	401-625	<400	0.0750	11-16-23	<20	TDC	TITUS
CG-E	18X18	24X24	626-900	<400	0.0750	13-20-28	<20	TDC	TITUS
CG-F	21X21	24X24	901-1150	<400	0.0750	14-21-30	<20	TDC	TITUS

NOTES:
1. CONTRACTOR SHALL COORDINATE GRILL FRAME AND BORDER WITH ARCHITECT
2. COORDINATE WITH ARCHITECT FOR ELEVATION OF GRILL
3. DATA SHOWN FOR A4 DISCHARGE TYPE (4-WAY)

ARCHITECTURAL SUPPLY AIR GRILLES SCHEDULE (CEILING MOUNTED)

TAG	NOMINAL DUCT SIZE (IN)	GRILL SIZE (IN.)	CFM RANGE	MAXIMUM DUCT VELOCITY FPM	PRESSURE LOSS (IN. WG)	THROW (FT)	NC LEVEL	MODEL	MANUFACTURER
AG-A	6ø	12X12	0-125	<700	0.049	1-3-5	<20	OMNI	TITUS
AG-B	8ø	12X12	126-225	<700	0.058	3-4-7	<20	OMNI	TITUS
AG-C	10ø	24X24	226-375	<700	0.082	5-7-13	<20	OMNI	TITUS
AG-D	12ø	24X24	376-550	<700	0.115	6-9-16	<20	OMNI	TITUS
AG-E	14ø	24X24	551-725	<700	0.155	7-10-17	<20	OMNI	TITUS
AG-F	15ø	24X24	726-800	<700	0.177	7-11-19	<20	OMNI	TITUS

NOTES:
1. CONTRACTOR SHALL COORDINATE GRILL FRAME AND BORDER WITH ARCHITECT
2. COORDINATE WITH ARCHITECT FOR ELEVATION OF GRILL

HORIZONTAL RETURN/EXHAUST AIR GRILLES SCHEDULE (SIDEWALL/SIDE DUCT MOUNTED)

TAG	NOMINAL DUCT SIZE (IN)	GRILL SIZE (IN.)	CFM RANGE	MAXIMUM DUCT VELOCITY FPM	PRESSURE LOSS (IN. WG)	NC LEVEL	MODEL	MANUFACTURER
RR-A	6X6	7¾X7¾	0-100	<600	0.102	<20	355R	TITUS
RR-B	10X6	11¾X7¾	101-200	<600	0.102	<20	355R	TITUS
RR-C	14X6	15¾X7¾	201-275	<600	0.102	<20	355R	TITUS
RR-D	18X6	19¾X7¾	276-350	<600	0.102	<20	355R	TITUS
RR-E	24X6	25¾X7¾	351-475	<600	0.102	<20	355R	TITUS
RR-F	30X6	31¾X7¾	475-600	<600	0.102	<20	355R	TITUS
RR-G	30X8	31¾X9¾	601-800	<600	0.102	<20	355R	TITUS
RR-H	30X12	31¾X13¾	801-1150	<600	0.102	<20	355R	TITUS

NOTES:
1. CONTRACTOR SHALL COORDINATE GRILL FRAME AND BORDER WITH ARCHITECT
2. COORDINATE WITH ARCHITECT FOR ELEVATION OF GRILL

VERTICAL RETURN/EXHAUST AIR GRILLES SCHEDULE (CEILING MOUNTED)

TAG	NOMINAL DUCT SIZE (IN)	GRILL SIZE (IN.)	CFM RANGE	MAXIMUM DUCT VELOCITY FPM	PRESSURE LOSS (IN. WG)	NC LEVEL	MODEL	MANUFACTURER
CR-A	6X6	7¾X7¾	0-100	<600	0.102	<20	355R	TITUS
CR-B	8X8	9¾X9¾	101-200	<600	0.102	<20	355R	TITUS
CR-C	10X10	11¾X11¾	201-325	<600	0.102	<20	355R	TITUS
CR-D	12X12	13¾X13¾	325-475	<600	0.102	<20	355R	TITUS
CR-E	14X14	15¾X15¾	476-650	<600	0.102	<20	355R	TITUS
CR-F	16X16	17¾X17¾	651-850	<600	0.102	<20	355R	TITUS

NOTES:
1. CONTRACTOR SHALL COORDINATE GRILL FRAME AND BORDER WITH ARCHITECT
2. COORDINATE WITH ARCHITECT FOR ELEVATION OF GRILL

HORIZONTAL DISCHARGE SUPPLY AIR GRILLES SCHEDULE (SIDEWALL/SIDE DUCT MOUNTED)

TAG	NOMINAL DUCT SIZE (IN)	GRILL SIZE (IN.)	CFM RANGE	MAXIMUM DUCT VELOCITY FPM	PRESSURE LOSS (IN. WG)			THROW (FT)			NC LEVEL	MODEL	MANUFACTURER
					0°	22.5°	45°	0°	22.5°	45°			
SR-A	6X6	7¾X7¾	0-125	<700	0.090	0.100	0.152	10-14-22	8-11-15	4-6-9	<20	300RL	TITUS
SR-B	10X6	11¾X7¾	126-200	<600	0.066	0.074	0.111	11-17-24	9-13-19	5-7-11	<20	300RL	TITUS
SR-C	14X6	15¾X7¾	201-275	<600	0.066	0.074	0.111	13-20-28	10-15-22	6-9-13	<20	300RL	TITUS
SR-D	18X6	19¾X7¾	276-350	<600	0.066	0.074	0.111	15-23-33	12-18-25	7-10-15	<20	300RL	TITUS
SR-E	24X6	25¾X7¾	351-450	<600	0.066	0.074	0.111	18-27-39	14-21-30	8-12-17	<20	300RL	TITUS
SR-F	30X6	31¾X7¾	451-600	<600	0.066	0.074	0.111	20-30-43	16-23-34	9-14-19	<20	300RL	TITUS
SR-G	30X8	31¾X9¾	601-750	<600	0.066	0.074	0.111	23-35-50	18-27-39	10-16-23	<20	300RL	TITUS
SR-H	36X8	28¾X9¾	751-900	<500	0.046	0.051	0.077	26-38-55	20-30-43	12-17-25	<20	300RL	TITUS

NOTES:
1. CONTRACTOR SHALL COORDINATE GRILL FRAME AND BORDER WITH ARCHITECT
2. COORDINATE WITH ARCHITECT FOR ELEVATION OF GRILL

ELECTRIC CABINET UNIT HEATER SCHEDULE

"QMARK" AS STD

UNIT NO.	AREA SERVED	LOCATION	TYPE	AIR TEMP (°F)		HEATING COIL		FAN MOTOR			MFR. & MODEL NO.	REMARKS
				ENT.	LVG.	TOTAL BTU/HR	ELECT. kW	HP	RPM	Volts/Ph/Hz		
ECUH-1	FUEL OIL	SUB-CELLAR	WALL MOUNTED	50	-	3,410	1	-	-	120/1/60	SED1012	SEDSM ACCESSORY SURFACE MOUNTING

ENERGY RECOVERY VENTILATOR (ERV)

UNIT TAG	LOCATION	CAPACITY (CFM)	STATIC PRESSURE (IN WC)	ELECTRICAL DATA					MANUFACTURER	MODEL NUMBER	NOTES
				VOLTS	PHASE	HZ	MCA	MOP			
ERV-1	ROOF	660	1.25	208	1	60	11.3	15	RENEWAIRE	HE 10RT	

AHRI CERTIFIED® www.ahridirectory.org
Air-to-Air ERV
AHRI Standard 1060
Energy Recovery COMPONENT is certified. Actual performance in packaged equipment may vary.

ERV-1 DATA

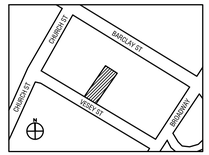
	SUMMER			WINTER		
	Outdoor Air	**Return Air**	**Supply Air**	**Outdoor Air**	**Return Air**	**Supply Air**
Standard Flow Rate **SCFM**	696*	620	660	696*	620	660
Actual Flow Rate **ACFM**	728*	634	680	620*	626	643
Dry Bulb **°F**	85.7	75.0	78.2	8.7	70.0	51.6
Wet Bulb **°F**	72.3	62.5	67.3	5.9	51.4	39.8
Enthalpy (H) **BTU/lb**	36.0	28.2	31.8	2.6	21.0	15.1
Moisture Ratio (MR) **grains/lb**	98.3	65.0	83.1	3.6	27.2	17.7
Supply Air - External Static Pressure **in w.g.**		1.25			1.25	
Exhaust Air - External Static Pressure **in w.g.**		1.25			1.25	
Sensible effectiveness **%**		74.6			74.6	
Total effectiveness **%**		57.3			72.2	
Enthalpy Recovery Ratio (ERR) **%** - 90.1 Compliance ✓		53.9			67.8	
Sensible Energy Recovery Ratio **%**		70.0			70.0	
Moisture removed **grains/lb**		15.2			-14.0	
	Sen	**Lat**	**Tot**	**Sen**	**Lat**	**Tot**
Original load **BTUH [Tons]**	7627 [0.6]	15656 [1.3]	23283 [1.9]	43695	10975	54669
Load with RenewAire **BTUH [Tons]**	2285 [0.2]	8459 [0.7]	10744 [0.9]	13090	4486	17577
Total energy saved **BTUH [Tons]**	5342 [0.4]	7197 [0.6]	12539 [1.0]	30604	6488	37093

*Note: OA Flow Rate values are gross airflow, all others are net airflow.
Note: For full certified ERV performance, please see AHRI 1060 Report.
Note: Sensible cooling design conditions were used for the summer performance results.

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SITE PLAN



DoB STAMPS:

No.	DATE	REVISION
	01/08/26	BID SET

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PROJECT:

18 VESEY STREET
MANHATTAN, NY
BLOCK: 88 LOT: 4

DRAWING TITLE:

MECHANICAL EQUIPMENT SCHEDULE II

DATE: 01/06/2026
PROJECT No.: 25086
CHECKED BY:
DRAWN BY: PY/LZ/BL
SEAL AND SIGNATURE

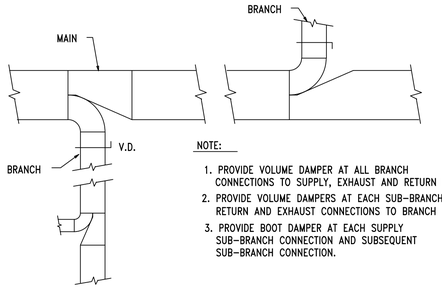
DWG. No.:
M402.00

N.Y.C D.O.B No. :
M01244603-S1

INTERPRETATION OF SINGLE LINE DUCTWORK

NOTE: WHERE DUCTWORK IS SHOWN IN SINGLE LINE, THE FOLLOWING DETAIL SHALL APPLY FOR ACTUAL CONSTRUCTION

SINGLE LINE	ACTUAL CONSTRUCTION
TEE CONVERGING/ DIVERGING	VOLUME DAMPER; SINGLE TURNING VANES
ELBOW	SINGLE BLADE TURNING VANE (EITHER TYPE ELBOW AT CONTR. OPTION–SUBJECT TO SPACE CONDITIONS). VANES SHALL BE WELDED TO RUNNERS.
(BRANCH AIR FLOW 25% OR HIGHER) SPLIT OR TAKE–OFF	A,B SHALL BE PROPORTIONAL TO AIR QUANTITY, 4" (10cm) MIN. IF A or B IS LESS THAN 4" (10cm), USE TAP DETAIL. VOLUME DAMPER IN SHORTER BRANCH. SQUARE THROAT ELBOW OPTION. D_2=4"(102mm) MIN. D_3=4"(102mm) MIN.
(BRANCH TAP 25% OR OR LESS AIR FLOW) TAP TAKE–OFF	SIDE TAP BRANCH SHALL BE LIMITED TO 25% OF MAIN AIR FLOW. SUPPLY. RETURN.
RISE OR DROP	OFFSET IN VERTICAL PLANE SHALL BE MADE WITH SMOOTH FITTINGS. ELEVATION
HORIZONTAL OFFSET	OFFSET IN HORIZONTAL PLANE SHALL BE MADE WITH SMOOTH FITTINGS.
PLENUM OVER DIFFUSER; CONN. TO LINEAR DIFFUSER PLENUM	PLENUM OVER LINEAR SUPPLY OR RETURN DIFFUSER OR GRILLE. CONICAL FITTINGS. REGULATION OF VOLUME DAMPER SHALL BE ACCESSIBLE THROUGH THE FACE OF DIFFUSER
FSM/F/SM FIRE RATED WALL; FIRE & SMOKE DAMPER	ACCESS DOOR IN DUCT. FSM–F/SM MOUNTING TYPE DEPENDS ON AIR VELOCITIES AND DUCT SIZE – SEE SPECIFICATIONS.
HTG. COIL; HEATING COIL	FLARE/TRANSITION DUCT TO ACCOMMODATE COIL PER LATEST SMACNA STANDARD. COIL WITH FLANGED CONNECTION ALL AROUND. ACCESS DOORS. AIR FLOW

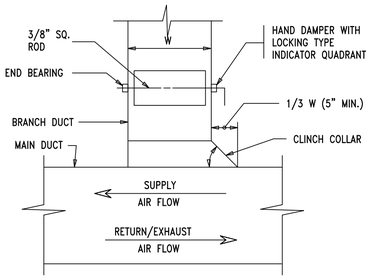


NOTE:

1. PROVIDE VOLUME DAMPER AT ALL BRANCH CONNECTIONS TO SUPPLY, EXHAUST AND RETURN MAINS.
2. PROVIDE VOLUME DAMPERS AT EACH SUB–BRANCH RETURN AND EXHAUST CONNECTIONS TO BRANCH
3. PROVIDE BOOT DAMPER AT EACH SUPPLY SUB–BRANCH CONNECTION AND SUBSEQUENT SUB–BRANCH CONNECTION.

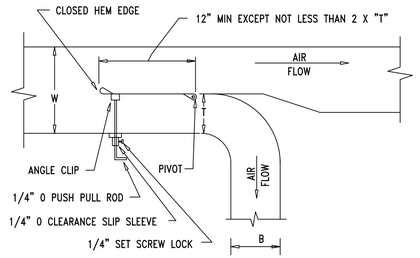
DUCT CONNECTIONS: BRANCH + SUB-BRANCH

NOT TO SCALE



RECTANGULAR DUCT TAP WITH VOLUME DAMPER

NOT TO SCALE:

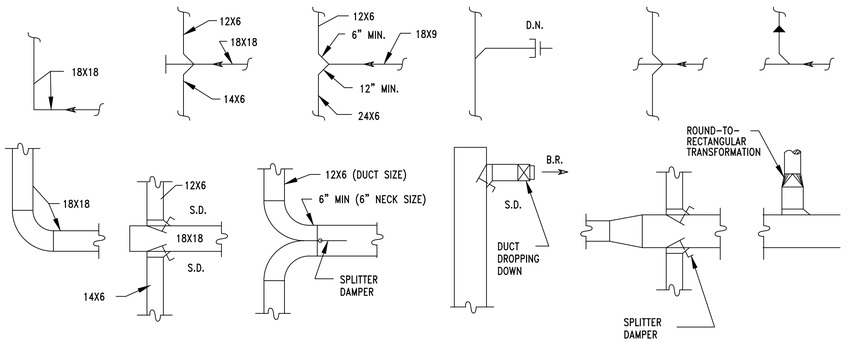


NOTES:

1. FURNISH THIS TYPE CONNECTION WHERE NOTED AND WHEN SINGLE LINE DUCTWORK IS INDICATED AS THIS FOR LOW PRESSURE BRANCHES WITH MORE THAN 700 CFM AND MEDIUM PRESSURE BRANCHES WITH MORE THAN 1000 CFM
2. ONE ROD UP TO 24" DEPTH
3. TWO RODS 25" UP TO 36"
4. OVER 36" USE DUCT BRANCH WITH THROAT AND VOLUME DAMPER
5. $T = \frac{CFM \quad B \times W}{CFM \quad W}$, MINIMUM 4 INCHES

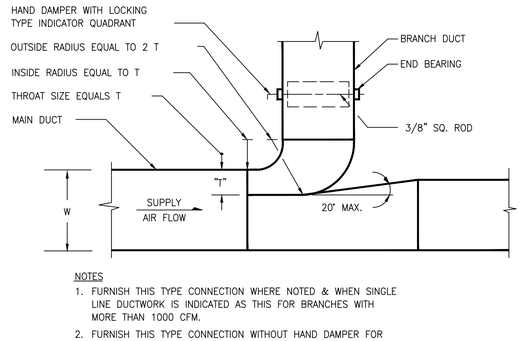
RECTANGULAR DUCT BRANCH WITH THROAT + SPLITTER DAMPER

NOT TO SCALE



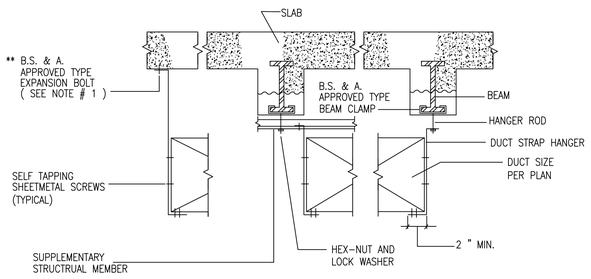
TYPICAL SINGLE LINE DUCT DETAILS

NOT TO SCALE



NOTES

1. FURNISH THIS TYPE CONNECTION WHERE NOTED & WHEN SINGLE LINE DUCTWORK IS INDICATED AS THIS FOR BRANCHES WITH MORE THAN 1000 CFM.
2. FURNISH THIS TYPE CONNECTION WITHOUT HAND DAMPER FOR ALL HIGH PRESSURE DUCTWORK BRANCHES.
3. MUST BE USED WHEN W $\geq$ 36".
4. MINIMUM T = 4".



NOTES:

1. THE USE OF EXPANSION BOLTS IN CINDER CONCRETE IS PROHIBITED.

** BOARD OF STANDARD AND APPEALS

HANGER SIZE

DUCT SIZE	HANGER SIZE
TO 2 SQ.FT.	1"X 1/16
OVER 2 SQ.FT.	1"X 1/8

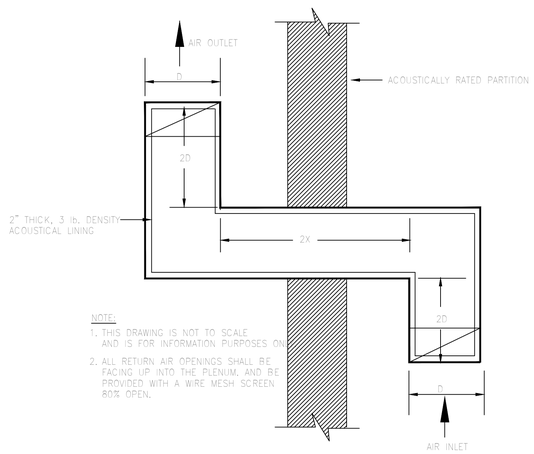
HANGER SPACING

DUCT SIZE	MAX. SPACING
TO 4 SQ.FT.	8'–0" O. C.
TO 6 SQ.FT.	6'–0" O. C.

RECTANGULAR DUCT SUPPORT DETAIL

NO SCALE

[TO BE USED FOR DUCTWORK WHOSE CONSTRUCTION CLASSIFICATION IS UNDER 2 "W.G., WITH WIDTHS UP TO 42 INCHES AND CROSS SECTIONAL AREA LESS THAN 6 SQ. FT.]

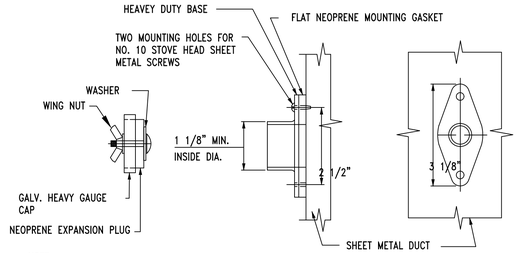


NOTE:

1. THIS DRAWING IS NOT TO SCALE AND IS FOR INFORMATION PURPOSES ONLY.
2. ALL RETURN AIR OPENINGS SHALL BE FACING UP INTO THE PLENUM, AND BE PROVIDED WITH A WIRE MESH SCREEN 80% OPEN.

TRANSFER DUCT DETAIL

RENTANGULAR DUCT BRANCH WITH THROAT & VOLUME DAMPER



NOTE:

INSTALL ONE TEST HOLE FITTING ON BOTH (OPPOSITE) SIDES OF DISCHARGE DUCT FROM ALL SUPPLY UNITS AND EXHAUST UNITS AND OTHER LOCATIONS SHOWN ON PLANS.

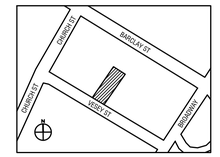
DETAIL OF TEST FITING FOR AIR READINGS

NOT TO SCALE

ALL CITY ENGINEERING, P.C.
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PROJECT:

18 VESEY STREET
MANHATTAN, NY
BLOCK: 88 LOT: 4

DRAWING TITLE:

MECHANICAL DETAILS

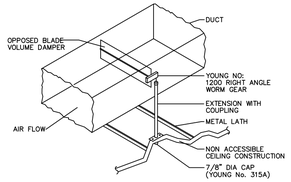
DATE:	01/06/2026
PROJECT No.:	25086
CHECKED BY:	
DRAWN BY:	PY/LZ/BL

SEAL AND SIGNATURE

DWG. No.:
M501.00

Sheets: 13 of 18

N.Y.C D.O.B No. :
M01244603–S1

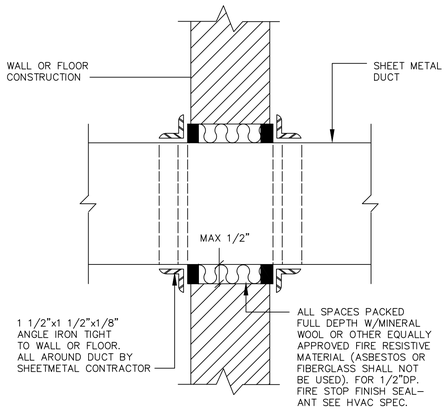


NOTES:

1. WHEN DAMPER IS LOCATED WITHIN 10' OF A DIFFUSER, OR WITHIN A LOBBY OR SPECIALTY SPACE (COORDINATE WITH ARCHITECT), VOLUME DAMPER ASSEMBLY SHALL BE OPERABLE THROUGH FACE OF NEAREST DIFFUSER, BY MEANS OF FLEXIBLE-CABLE BASED ON BOWDEN REMOTE CABLE CONTROL SYSTEM WITH YOUNG REGULATOR.
2. ALL OTHER NON-ACCESIBLE CEILING AREAS SHALL BE AS INDICATED.

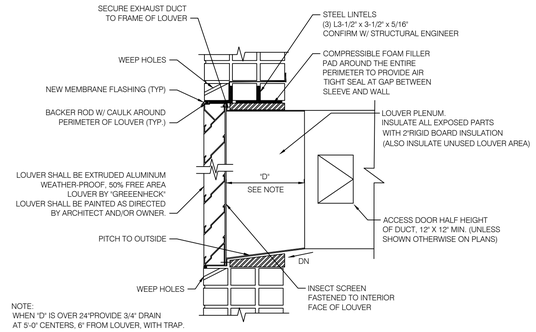
VOLUME CONTROL DAMPER

FOR NON ACCESSIBLE CEILINGS ONLY



NOTE:
1. FOR WALLS REQUIRING FIRE DAMPERS, SEE OTHER DETAILS.

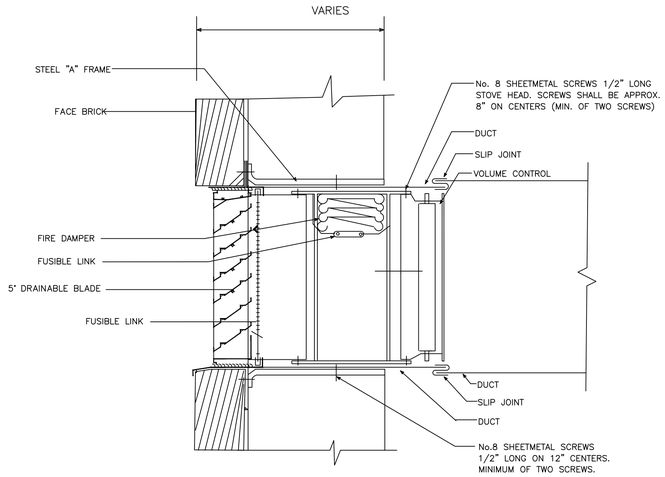
DETAIL OF CAULKING OF DUCT PIERCING WALLS OR FLOORS WHERE FIRE DAMPERS ARE NOT REQUIRED



NOTE:
WHEN "D" IS OVER 24"PROVIDE 3/4" DRAIN AT 5'-0" CENTERS, 6" FROM LOUVER, WITH TRAP.

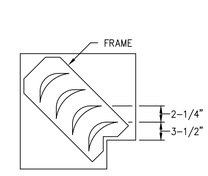
DUCTWORKS CONNECT TO LOUVER DETAIL

NO SCALE

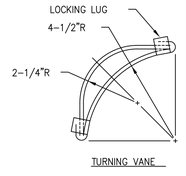


DUCTWORKS CONNECT TO LOUVER WITH FIRE DAMPER DETAIL

NO SCALE



VANED ELBOW

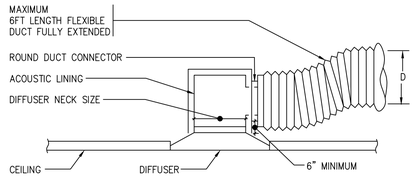


TURNING VANE

NOTES:

1. LOCKING LUGS INTEGRAL WITH VANE
2. MAXIMUM UNSUPPORTED VANE LENGTH 48"
3. FRAMES – BOLTED OR RIVETED TO ELBOW
4. VANES AND FRAMES – SAME GAUGE AS ELBOW

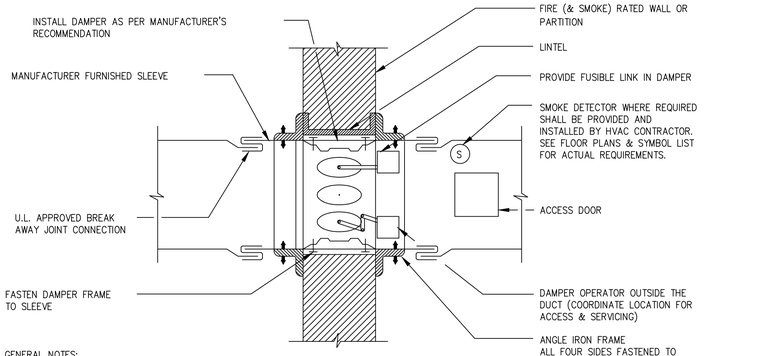
DOUBLE THICKNESS TURNING VANES FOR SQUARE ELBOW



SIDE CONNECTION

TOP CONNECTION

DIFFUSER CONNECTION DETAIL



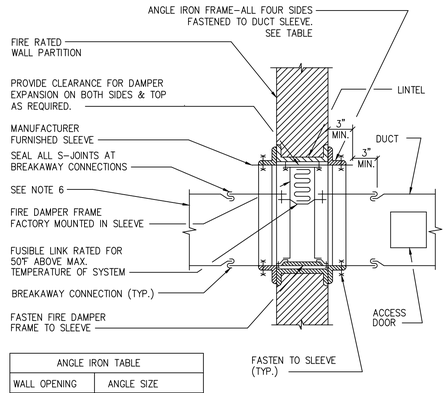
GENERAL NOTES:

1. DAMPER TO BE U.L. LABELED N.F.P.A. 90A AND BSA APPROVED
2. N.F.P.A. APPROVED INSTALLATION DETAILS TO BE PART OF SUBMISSION OF COMBINATION FIRE & SMOKE DAMPER FOR APPROVAL, WHICH SHALL MEET N.F.P.A. STANDARD 90A AND BSA APPROVED.
3. DETAILS SHOWN ARE FOR COMBINATION FIRE AND SMOKE DAMPERS IN HORIZONTAL DUCTWORK. FOR DAMPERS IN VERTICAL DUCTWORK, DETAIL IS SIMILAR.
4. ACCESS DOOR IS SHOWN IN SIDE OF DUCT.
5. DETAIL FOR SMOKE DAMPER SHALL BE SIMILAR TO COMBINATION DAMPER EXCEPT DELETE FUSIBLE LINK IN DAMPER.
6. THIS DETAIL IS FOR GUIDE ONLY. INSTALL DAMPER IN ACCORDANCE WITH MANUFACTURER'S RECOMMENDATIONS.

ANGLE IRON TABLE	
WALL OPENING	ANGLE SIZE
UP TO 54"	1 1/2"x1 1/2"x1/8"
55" TO 84"	3"x2"x 3/16"
85" TO 120"	3"x2"x 3/16"

NOTE: RETAINING ANGLES MUST LAP STRUCTURAL OPENING 1" MIN. AND COVER CORNERS OF OPENINGS

COMBINATION FIRE AND SMOKE DAMPER WITH FIRE'STAT OR RESETABLE LINK



ANGLE IRON TABLE	
WALL OPENING	ANGLE SIZE
UP TO 54"	1 1/2"x1 1/2"x1/8"
55" TO 84"	3"x2"x 3/16"
85" TO 120"	3"x2"x 3/16"

NOTE: RETAINING ANGLES MUST LAP STRUCTURAL OPENING 1" MIN. AND COVER CORNERS OF OPENINGS

GENERAL NOTES:

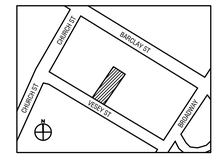
1. FIRE DAMPER TO BE U.L. LABELED N.F.P.A. 90 A.
2. N.F.P.A. APPROVED INSTALLATION DETAILS TO BE PART OF SUB-MISSION OF FIRE DAMPER FOR APPROVAL, WHICH SHALL MEET N.F.P.A. STANDARD 90 A.
3. DETAILS SHOWN ARE FOR FIRE DAMPERS IN HORIZONTAL DUCT-WORK FOR FIRE DAMPERS IN VERTICAL DUCTWORK DETAILS SIMILAR EXCEPT DAMPERS SHALL BE SPRING LOADED.
4. ACCESS DOOR IS SHOWN IN SIDE OF DUCT; IF FUSIBLE LINK IS MORE ACCESSIBLE FROM BOTTOM OF DUCT RELOCATE ACCESS DOOR.
5. U.L. APPROVED BREAKAWAY SLIP JOINT CONNECTION SHALL BE USED.
6. FIRE DAMPER FOR DUCT HEIGHT LARGER THAN 24 IN. CAN BE IN AIR STREAM.
7. THE DETAIL IS FOR GUIDE ONLY INSTALL FIRE DAMPER IN ACCORDANCE WITH MANUFACTURER'S RECOMMENDATION.

FIRE DAMPER DETAIL

ALL CITY ENGINEERING, P.C.
33 Great Neck Road #3
Great Neck, NY 11021
(516)321-9490
info@allcityengineering.com

OWNER:
SEAFARERS INTERNATIONAL HOUSE
123 EAST 15TH STREET
NEW YORK, NY 1003
TEL: (212) 677-4800
EMAIL: LUCAS@SIHNYC.ORG

SITE PLAN



DoB STAMPS:

No.	DATE	REVISION
	01/08/26	BID SET

NOTE: Drawing may be printed at reduced scale

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PROJECT:

18 VESEY STREET
MANHATTAN, NY
BLOCK: 88 LOT: 4

DRAWING TITLE:

MECHANICAL DETAILS II

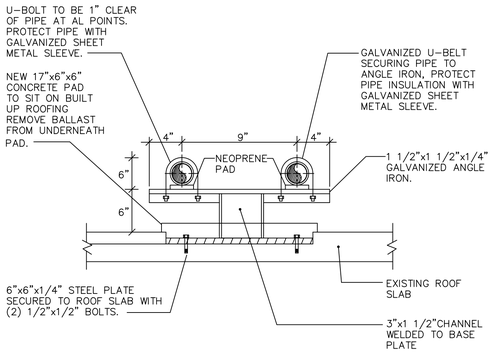
DATE:	01/06/2026
PROJECT No.:	25086
CHECKED BY:	
DRAWN BY:	PY/LZ/BL

SEAL AND SIGNATURE

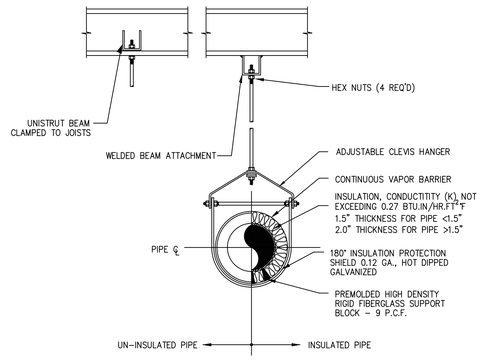
DWG. No.:
M502.00

Sheets: 14 of 18

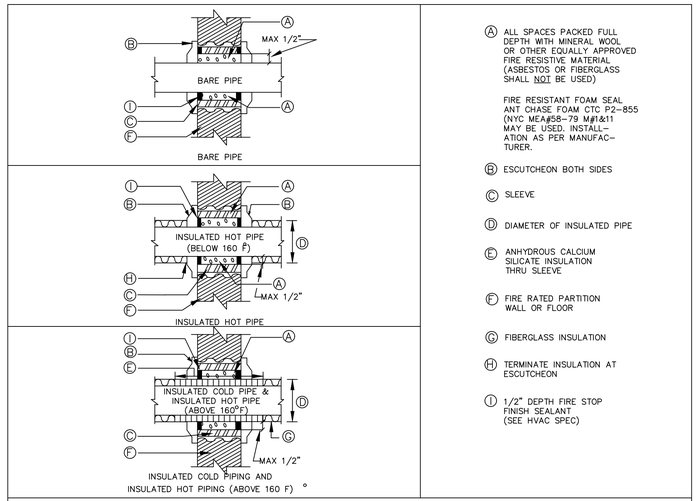
N.Y.C D.O.B No. :
M01244603-S1



PIPES SUPPORT ON ROOF DETAIL



CLEVIS HANGER



DETAIL OF PIPING PIERCING FIRE RATED PARTITIONS, WALLS AND FLOORS

PIPING PENETRATED WALL

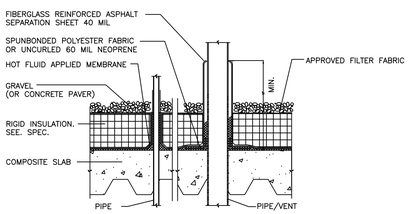
NOT TO SCALE

DETAIL OF REFRIGERATION PIPE ENCLOSURE

N.T.S.

PIPE PENETRATION – SLAB ON GRADE AND WATERPROOF SLAB

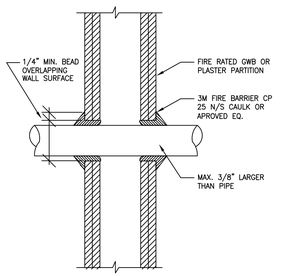
N.T.S.



NOTES: SOME PIPE MATERIALS (PVC, COPPER, BRASS) MAY REQUIRE ROUGHENING/SANDING, IN ADDITION TO PRIMING W/SURFACE CONDITIONER, FOR PROPER ADHESION OF THE APPLIED MEMBRANE. METAL PIPES MUST BE FREE OF ALL OIL AND RUST.

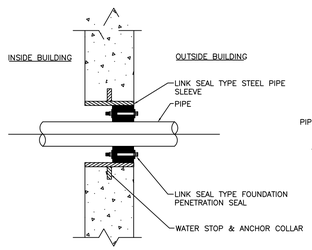
PIPE PENETRATION DETAIL

N.T.S.



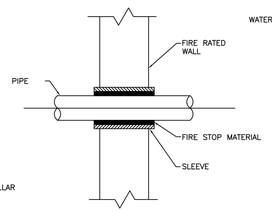
PIPE PENETRATION IN FIRE RATED GWB OR PLASTER PARTITION

N.T.S.



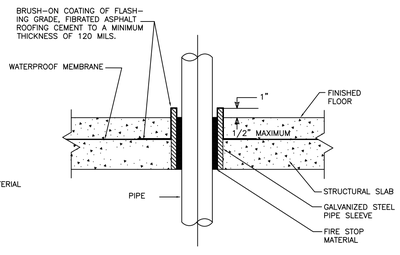
PIPE PENETRATION THRU FOUNDATION WALL

N.T.S.



PIPE PENETRATION THRU FIRE RATED WALL

N.T.S.



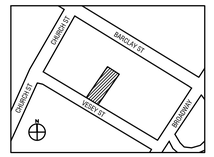
PIPE PENETRATION THRU WATERPROOF SLAB

N.T.S.

ACE
ALL CITY ENGINEERING, P.C.
33 Great Neck Road #3
Great Neck, NY 11021
(516)321-9490
info@allcityengineering.com

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NEW YORK, NY 1003
TEL: (212) 677-4800
EMAIL: LUCAS@SIHNYC.ORG

SITE PLAN



DoB STAMPS:

No.	DATE	REVISION
	01/08/26	BID SET

NOTE: Drawing may be printed at reduced scale

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PROJECT:

18 VESEY STREET
MANHATTAN, NY
BLOCK: 88 LOT: 4

DRAWING TITLE:

MECHANICAL DETAILS III

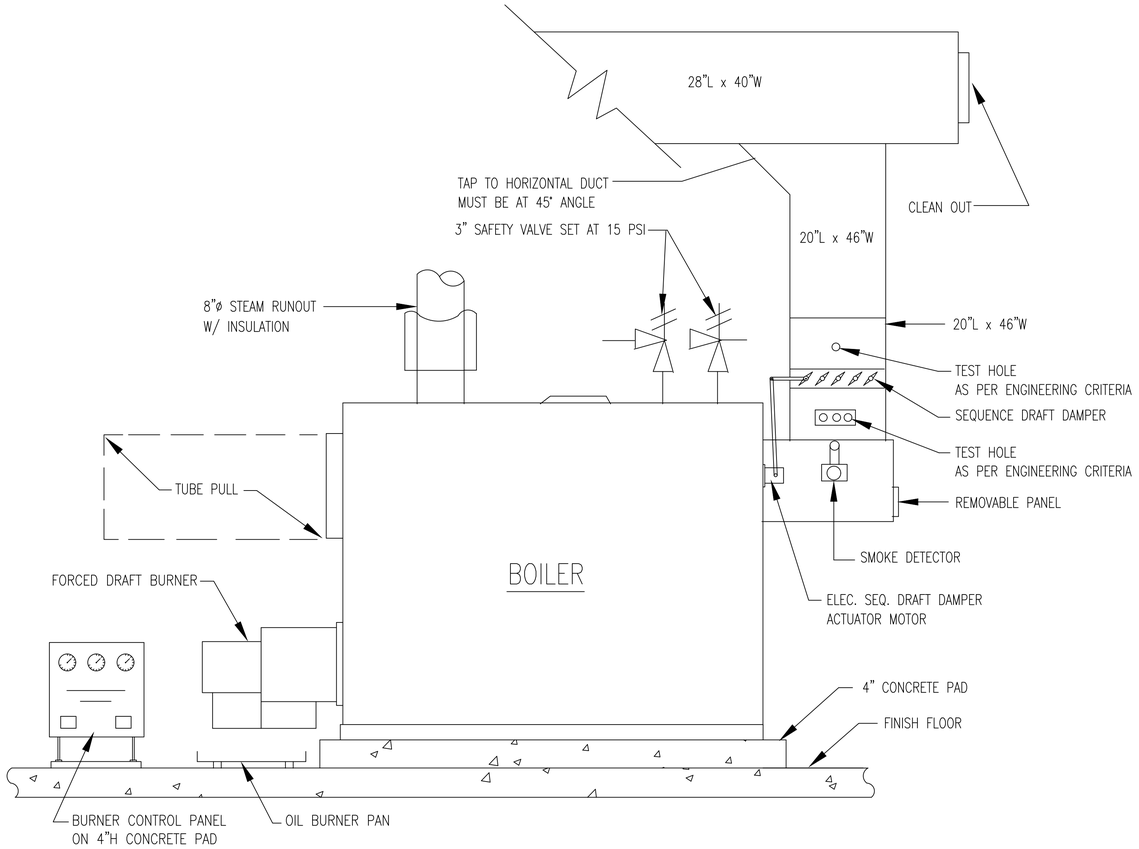
DATE:	01/06/2026
PROJECT No.:	25086
CHECKED BY:	
DRAWN BY:	PY/LZ/BL

SEAL AND SIGNATURE

DWG. No.:
M503.00

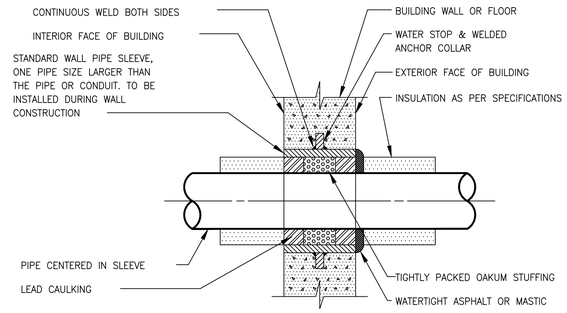
Sheets: 15 of 18

N.Y.C D.O.B No. :
M01244603-S1



TYPICAL BOILER ELEVATION

NOT TO SCALE



NOTES:

1. WHEN SEALING FLOOR PENETRATIONS, EXTEND SLEEVE 3" ABOVE FINISHED FLOOR.
2. IN EXISTING CONSTRUCTION, A CORE DRILL ONE PIPE SIZE LARGER THAN THE PIPE OR CONDUIT MAY BE USED IN LIEU OF A SLEEVE.

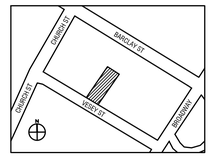
CAULKED SEAL ASSEMBLY

(PIPE PENENTRATION THROUGH WALL)

ALL CITY ENGINEERING, P.C.
33 Great Neck Road #3
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(516)321-9490
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TEL: (212) 677-4800
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SITE PLAN



DoB STAMPS:

No.	DATE	REVISION
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PROJECT:

18 VESEY STREET
MANHATTAN, NY
BLOCK: 88 LOT: 4

DRAWING TITLE:

MECHANICAL DETAILS IV

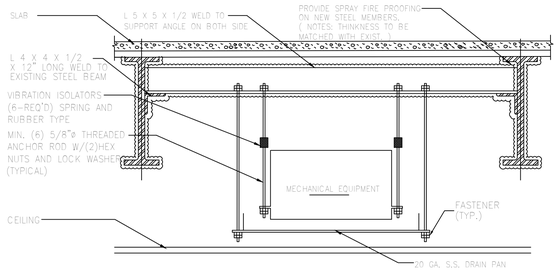
DATE:	01/06/2026
PROJECT No.:	25086
CHECKED BY:	
DRAWN BY:	PY/LZ/BL

SEAL AND SIGNATURE

DWG. No.:
M504.00

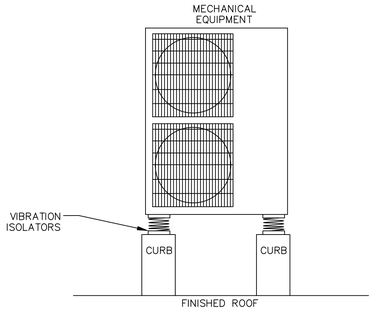
Sheets: 16 of 18

N.Y.C D.O.B No. :
M01244603-S1

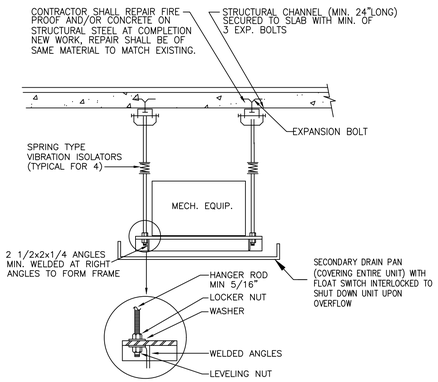


MECHANICAL EQUIPMENT SUPPORT DETAIL

NOT TO SCALE

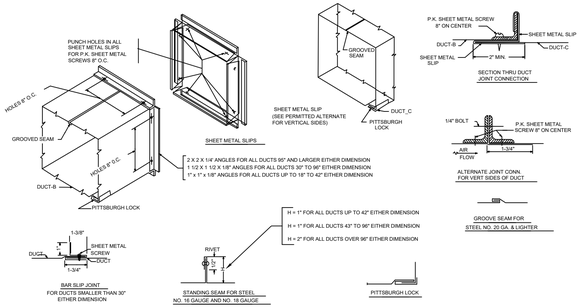


CURB MOUNTED MECHANICAL UNIT DETAIL



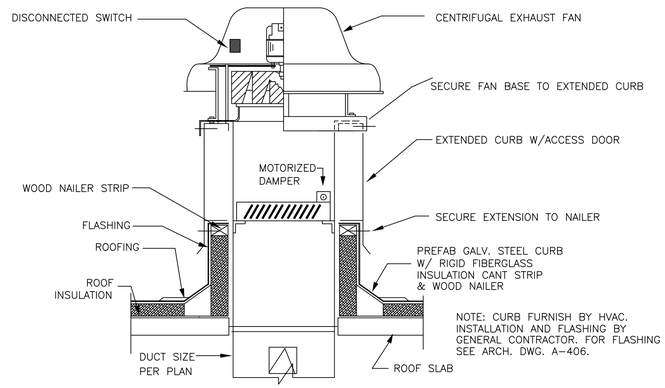
MECHANICAL AC UNIT SUPPORT DETAIL

N.T.S.



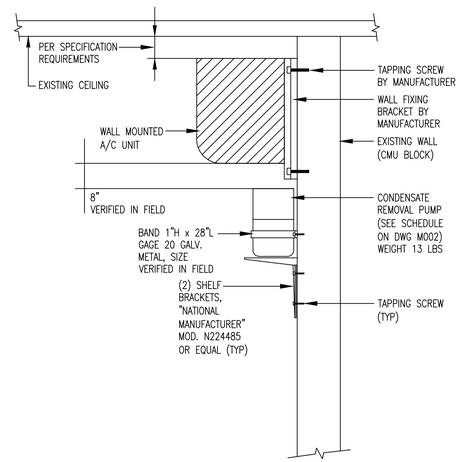
METHOD OF CONNECTING DUCT JOINTS

NO SCALE



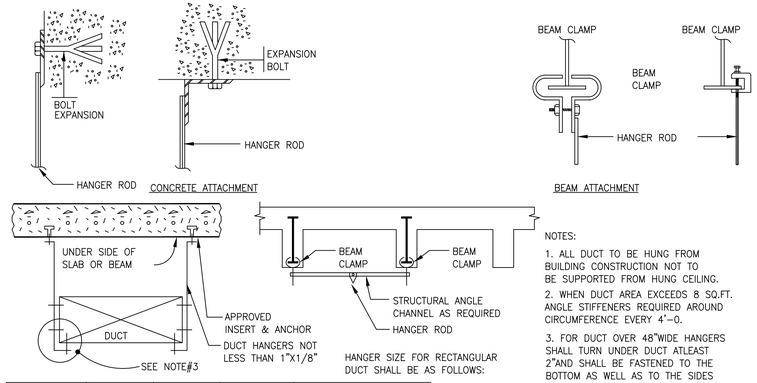
TYPICAL CURB MOUNTED ROOF EXHAUST FAN

NO SCALE



TYPICAL SUPPORT FOR WALL MOUNTED A/C UNITS AND CONDENSATE REMOVAL PUMP

NTS



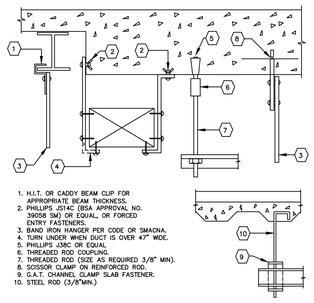
HANGER SIZE FOR RECTANGULAR DUCT SHALL BE AS FOLLOWS:

LONGEST DIMENSION OF DUCT	ROUND HANGERS	MAX SPACING SEE NOTE #4	STRAP HANGERS	TRAPEZE SHELF ANGLES
UP TO 18"	8"GA.WIRE	8'-0"	1"X1/8"	1"X1"X 1/8"
19" TO 30"	1/4" ROD	8'-0"	1"X1/8"	1"X1"X 1/8"
31" TO 42"	1/4" ROD	8'-0"	1"X1/8"GA.	1 1/2"X 1 1/2"X 1/8"
43" TO 60"	3/8" ROD	4'-0"	———	1 1/2"X 1 1/2"X 1/8"
61" TO 84"	3/8" ROD	4'-0"	———	2"X2"X1/8"
85" TO 96"	3/8" ROD	4'-0"	———	2"X2"X 3/16"
OVER 97"	3/8" ROD	4'-0"	———	2"X2"X 1/4"

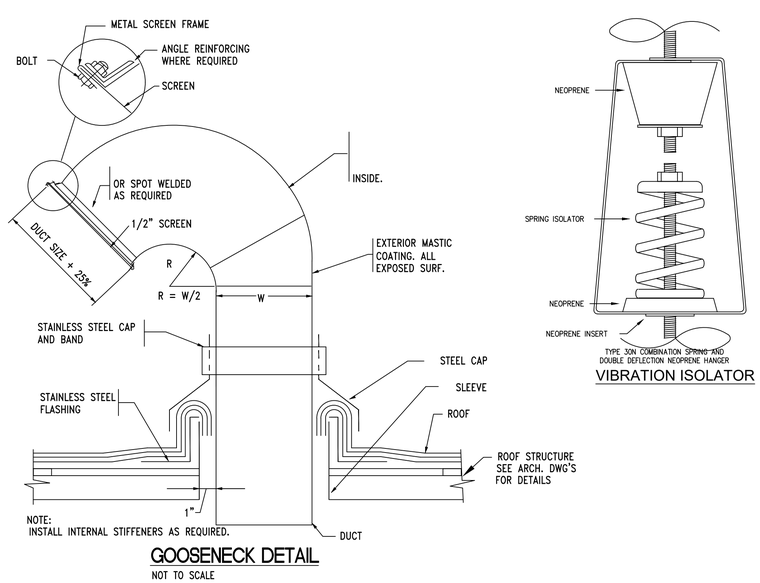
NOTES:

1. ALL DUCT TO BE HUNG FROM BUILDING CONSTRUCTION NOT TO BE SUPPORTED FROM HUNG CEILING.
2. WHEN DUCT AREA EXCEEDS 8 SQ.FT. ANGLE STIFFENERS REQUIRED AROUND CIRCUMFERENCE EVERY 4'-0.
3. FOR DUCT OVER 48"WIDE HANGERS SHALL TURN UNDER DUCT ATLEAST 2"AND SHALL BE FASTENED TO THE BOTTOM AS WELL AS TO THE SIDES OF THE DUCT.
4. FOR DUCTS WITH A CROSS SECTIONAL AREA OF 4 SQ.FT. OR LESS. HANGERS SHALL BE NO MORE THAN 8FT. A PART. FOR DUCTS WITH A CROSS SECTIONAL AREA OF MORE THAN 4 SQ.FT. BUT NOT OVER 8 SQ.FT. HANGERS SHALL BE NOT MORE THAN 6 FT. A PART. AND FOR DUCTS WITH A CROSS SECTIONAL AREA OF MORE THAN 8 SQ.FT. HANGERS SHALL BE NOT MORE THAN 4 FT. A PART. THE DISTANCES BETWEEN SHALL BE MEASURED LINEARLY ALONG THE DUCT.

DUCT SUPPORT DETAIL



METHOD OF HANGING DUCTWORK



GOOSENECK DETAIL

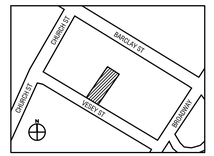
NOT TO SCALE

VIBRATION ISOLATOR

ALL CITY ENGINEERING, P.C.
33 Great Neck Road #3
Great Neck, NY 11021
(516)321-9490
info@allcityengineering.com

OWNER:
SEAFARERS INTERNATIONAL HOUSE
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SITE PLAN



DoB STAMPS:

No.	DATE	REVISION
	01/08/26	BID SET

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PROJECT:

18 VESEY STREET
MANHATTAN, NY
BLOCK: 88 LOT: 4

DRAWING TITLE:

MECHANICAL DETAILS V

DATE:	01/06/2026
PROJECT No.:	25086
CHECKED BY:	
DRAWN BY:	PY/LZ/BL

SEAL AND SIGNATURE

DWG. No.:
M505.00

Sheets: 17 of 18

N.Y.C D.O.B No. :
M01244603-S1

SPECIFICATIONS

1. GENERAL

1.A. THE "GENERAL CONDITIONS OF THE CONTRACT FOR CONSTRUCTION," AIA DOCUMENT A201, LATEST EDITION, AND THESE SPECIFICATIONS AS APPLICABLE ARE PART OF THIS CONTRACT.

1.B. ALL APPLICABLE CODES, LAWS AND REGULATIONS GOVERNING OR RELATING TO ANY PORTION OF THIS WORK ARE HEREBY INCORPORATED INTO AND MADE A PART OF THESE SPECIFICATIONS, AND THEIR PROVISIONS SHALL BE CARRIED OUT BY THE CONTRACTOR WHO SHALL INFORM THE OWNER, PRIOR TO SUBMITTING A PROPOSAL, OF ANY WORK OR MATERIALS WHICH VIOLATE ANY OF THE ABOVE LAWS AND REGULATIONS. ANY WORK DONE BY THE CONTRACTOR CAUSING SUCH VIOLATION SHALL BE CORRECTED BY THE CONTRACTOR.

1.C. EACH SPACE IN WHICH EQUIPMENT MUST BE MOVED THROUGH, MUST BE INVESTIGATED. WHERE NECESSARY, EQUIPMENT SHALL BE SHIPPED FROM THE MANUFACTURER IN SECTIONS WHERE EACH SECTION SIZE IS SUITABLE FOR BEING MOVED THROUGH UNRESTRICTED SPACES. ASCERTAIN FROM BUILDING OWNER AT WHAT TIMES OF THE DAY EQUIPMENT MAY BE MOVED THROUGH SUCH AREAS.

1.D. DUCTWORK AND PIPING IS SHOWN DIAGRAMMATICALLY AND DOES NOT SHOW ALL OFFSETS, DROPS AND RISES OF RUNS. THE CONTRACTOR SHALL ALLOW IN HIS PRICE FOR ROUTING OF DUCTWORK AND PIPING TO AVOID OBSTRUCTIONS. EXACT LOCATIONS ARE SUBJECT TO APPROVAL OF ENGINEER. COORDINATION WITH THE EXISTING SERVICES, INCLUDING THOSE OF OTHER TRADES IS REQUIRED.

1.E. SUPPORT ALL DUCTWORK AND PIPING FROM BUILDING STRUCTURE AND/OR FRAMING IN AN APPROVED MANNER. WHERE OVERHEAD CONSTRUCTION DOES NOT PERMIT FASTENING OR SUPPORTS FOR EQUIPMENT, FURNISH ADDITIONAL FRAMING. INSERTS SHALL BE STEEL, SLOTTED TYPE AND FACTORY PAINTED. SINGLE ROD SHALL BE GRINNELL FIG. 281 OR SIMILAR. MULTI-ROD SHALL BE FEE & MASON SERIES 9000 WITH END CAPS AND CLOSURE STRIPS OR SIMILAR. MAXIMUM LOAD INCLUDING PIPES, DUCTWORK CONTENTS AND COVERING SHALL NOT EXCEED 75% OF RATED INSERT CAPABILITY. WHEN SUPPORTING FROM BUILDING USE BEAM CLAMPS IN APPROVED MANNER.

1.F. INSTALL WORK SO AS TO BE READILY ACCESSIBLE FOR OPERATION, MAINTENANCE AND REPAIR. MINOR DEVIATIONS FROM DRAWINGS MAY BE MADE TO ACCOMPLISH THIS, BUT CHANGES WHICH INVOLVE EXTRA COST SHALL NOT BE MADE WITHOUT APPROVAL.

1.G. THE CONTRACTOR SHALL KEEP ALL EQUIPMENT AND MATERIALS, AND ALL PARTS OF THE BUILDING, EXTERIOR SPACES, ADJACENT STREETS, SIDEWALKS AND PAVEMENTS, FREE FROM MATERIAL AND DEBRIS RESULTING FROM THE EXECUTION OF THIS WORK. EXCESS MATERIALS WILL NOT BE PERMITTED TO ACCUMULATE EITHER ON THE INTERIOR OR THE EXTERIOR.

1.H. SEAL OPENINGS AROUND DUCTS AND PIPING THROUGH PARTITIONS, WALLS AND FLOORS (NOT IN SHAFTS) WITH MINERAL WOOL OR OTHER NONCOMBUSTIBLE MATERIAL.

1.I. PROVIDE ALL NECESSARY FLASHING AND COUNTERFLASHING TO MAINTAIN THE WATERPROOFING INTEGRITY OF THIS BUILDING AS REQUIRED BY THE INSTALLATION OR REMOVAL OF PIPES, DUCTS, LOUVERS, CONDUIT, AND EQUIPMENT. PROVIDE EQUIPMENT CURBS AND DUNNAGE STEEL AS REQUIRED.

1.J. ALL PRESENT MATERIAL, EQUIPMENT AND CONSTRUCTION DEBRIS TO BE REMOVED UNDER THIS CONTRACT SHALL BECOME THE PROPERTY OF THE CONTRACTOR AND BE PROPERLY DISPOSED OF BY THEM WITH THE EXCEPTION OF SPECIFIC EQUIPMENT AND APPARATUS REQUESTED BY THE BUILDING REPRESENTATIVE, ARCHITECT OR AS NOTED TO BE RELOCATED ON THE DRAWINGS.

1.K. THE WORK IN THE BUILDING SHALL BE DONE WHEN AND AS DIRECTED, AND IN A MANNER SATISFACTORY TO THE OWNER.

1.L. ALL EQUIPMENT SHALL HAVE AN MEA AND/OR BSA NUMBER. THIS INFORMATION MUST BE INCLUDED IN THE SUBMITTAL PACKAGE.

1.M. ALL MATERIAL AND EQUIPMENT TO BE NEW UNLESS OTHERWISE NOTED AND SHALL BE IN ACCORDANCE WITH BUILDING STANDARDS.

1.N. INSURANCE SHALL BE IN ACCORDANCE WITH BUILDING REQUIREMENTS AND INCLUDE A HARMLESS CLAUSE FOR OWNER AND ENGINEER

1.O. THE FINAL ACCEPTANCE WILL BE MADE AFTER THE CONTRACTOR HAS ADJUSTED HIS EQUIPMENT, BALANCED THE VARIOUS SYSTEMS, DEMONSTRATED THAT IT FULFILLS THE REQUIREMENTS OF THE DRAWINGS AND SPECIFICATIONS AND FURNISHED ALL THE REQUIRED CERTIFICATES OF INSPECTION AND APPROVAL.

1.P. SPECIFICATIONS ARE OF SIMPLIFIED FORM AND INCLUDE INCOMPLETE SENTENCES. WORDS OR PHRASES SUCH AS "THE CONTRACTOR SHALL," "SHALL BE," "FURNISH," "PROVIDE," "A," "THE," AND "ALL" HAVE BEEN OMITTED FOR BREVITY.

1.Q. DEFINITIONS

1.Q.1. "PROVIDE": TO SUPPLY, INSTALL AND CONNECT UP COMPLETE AND READY FOR SAFE AND REGULAR OPERATION THE PARTICULAR WORK REFERRED TO UNLESS SPECIFICALLY OTHERWISE NOTED.
1.Q.2. "INSTALL": TO ERECT, MOUNT AND CONNECT COMPLETE WITH RELATED ACCESSORIES.
1.Q.3. "FURNISH" OR "SUPPLY": TO PURCHASE, PROCURE, ACQUIRE AND DELIVER COMPLETE WITH RELATED ACCESSORIES.
1.Q.4. "WORK": LABOR, MATERIALS, EQUIPMENT, APPARATUS, CONTROLS, ACCESSORIES AND OTHER ITEMS REQUIRED FOR PROPER AND COMPLETE INSTALLATION.
1.Q.5. "CONCEALED": EMBEDDED IN MASONRY OR OTHER CONSTRUCTION, INSTALLED IN FURRED SPACES, WITHIN DOUBLE PARTITIONS OR HUNG CEILINGS, IN TRENCHES, IN CRAWL SPACES, OR IN ENCLOSURES.
1.Q.6. "EXPOSED": NOT INSTALLED UNDERGROUND OR "CONCEALED" AS DEFINED ABOVE.
1.Q.7. "SIMILAR" OR "EQUAL": EQUAL IN MATERIALS, WEIGHT, SIZE, DESIGN AND EFFICIENCY OF SPECIFIED PRODUCT.

2. SCOPE OF WORK

2.A. THE WORK UNDER CONTRACT INCLUDES ALL LABOR, MATERIALS AND APPLIANCES NECESSARY FOR THE FURNISHING, INSTALLING AND TESTING, COMPLETE AND READY FOR SAFE OPERATION OF THE SYSTEMS. WORK SHALL BE INSTALLED IN A NEAT, WORKMANLIKE MANNER.

2.B. THE CONTRACTOR SHALL GIVE NECESSARY NOTICE, FILE DRAWINGS AND SPECIFICATIONS WITH THE DEPARTMENT HAVING JURISDICTION, OBTAIN PERMITS OR LICENSES NECESSARY TO CARRY OUT THIS WORK AND PAY ALL FEES THEREFORE. THE CONTRACTOR SHALL ARRANGE FOR INSPECTION AND TESTS OF ANY OR ALL PARTS OF THE WORK IF SO REQUIRED BY AUTHORITIES AND PAY ALL CHARGES FOR SAME. THE CONTRACTOR SHALL PAY ALL COSTS FOR, AND FURNISH TO THE OWNER BEFORE FINAL BILLING, ALL CERTIFICATES NECESSARY AS EVIDENCE THAT THE WORK INSTALLED CONFORMS WITH ALL REGULATIONS WHERE THEY APPLY TO THIS WORK.

2.C. THE CONTRACTOR SHALL FURNISH A WRITTEN GUARANTEE TO REPLACE OR REPAIR PROMPTLY AND ASSUME RESPONSIBILITY FOR ALL EXPENSES INCURRED FOR ANY WORKMANSHIP AND EQUIPMENT IN WHICH DEFECTS DEVELOP WITHIN ONE YEAR FROM THE DATE OF FINAL CERTIFICATE FOR PAYMENT AND/OR FROM DATE OR ACTUAL USE OF EQUIPMENT OR OCCUPANCY OF SPACES, BY OWNER, INCLUDED UNDER THE VARIOUS PARTS OF THE WORK, WHICHEVER DATE IS EARLIER. THIS WORK SHALL BE DONE AS DIRECTED BY THE OWNER. THIS GUARANTEE SHALL ALSO PROVIDE THAT WHERE DEFECTS OCCUR, THE CONTRACTOR WILL ASSUME RESPONSIBILITY FOR ALL EXPENSES INCURRED IN REPAIRING AND REPLACING WORK OF OTHER TRADES AFFECTED BY DEFECTS, REPAIRS OR REPLACEMENTS IN EQUIPMENT SUPPLIED BY THE CONTRACTOR.

2.D. CONTROLLED INSPECTION BY A LICENSED PROFESSIONAL ENGINEER TO BE HIRED BY THIS CONTRACTOR.

2.E. PRIOR TO THE INSTALLATION OF ANY WORK AND PROCUREMENT OF EQUIPMENT PROVIDE COMPLETE SET OF COORDINATED SHOP DRAWINGS OF ALL NEW EQUIPMENT,DUCTWORK, PIPING AND CONTROL SYSTEMS INDICATING CAPACITY DIMENSIONS AND SEQUENCE OF OPERATION FOR WRITTEN APPROVAL BY THE AOR AND EOR.

3. SHOP DRAWINGS

3.A. INDICATE ON EACH SUBMISSION: PROJECT NAME AND LOCATION, ARCHITECT AND ENGINEER, ITEM IDENTIFICATION AND APPROVAL STAMP OF PRIME CONTRACTOR.

3.B. SUBMISSIONS:

3.B.1. SUBMISSIONS 11 IN. X 17 IN. OR SMALLER: IF THE SUBMISSION IS A CATALOG CUT, THEN THE CONTRACTOR SHALL SUBMIT ONE ORIGINAL AND TWO COPIES. OTHERWISE, HE SHALL SUBMIT THREE COPIES. THE ARCHITECT WILL FORWARD THE ORIGINAL AND ONE COPY (TWO COPIES WHEN NO ORIGINAL IS RECEIVED) TO THE ENGINEER. ALL CATALOG CUTS SHALL BE COMPLETE.

3.B.2. SUBMISSIONS LARGER THAN 11 IN. X 17 IN.: SUBMIT TWO PRINTS AND ONE PAPER SEPIA TO THE ARCHITECT. THE ARCHITECT WILL FORWARD ONE TO THE ENGINEER

3.C. SUBMIT SHOP DRAWINGS FOR THE FOLLOWING:

3.C.1. DUCTWORK LAYOUT AND SHEET METAL DESIGNS.
3.C.2. AIR OUTLETS.
3.C.3. AIR BALANCE REPORTS.
3.C.4. OPERATING SEQUENCES.
3.C.5. VIBRATION ISOLATION.
3.C.6. INSULATION.
3.C.7. AUTOMATIC CONTROL SYSTEMS AND DEVICES.

4. AS-BUILTS AND EQUIPMENT OPERATION INSTRUCTIONS

4.A. ON COMPLETION AND ACCEPTANCE OF WORK, THIS CONTRACTOR SHALL FURNISH WRITTEN INSTRUCTIONS, EQUIPMENT MANUALS AND DEMONSTRATE TO THE OWNER THE PROPER OPERATION AND MAINTENANCE OF ALL EQUIPMENT AND APPARATUS FURNISHED UNDER THIS CONTRACT.

4.B. THESE INSTRUCTIONS SHALL BE TYPED ON 8-1/2 IN. X 11 IN. PAPER AND BOUND IN THREE-RING BINDERS WITH CLEAR ACETATE COVERS. THE CONTRACTOR SHALL GIVE THREE COPIES OF THE INSTRUCTIONS TO THE OWNER AND ONE COPY TO THE ENGINEER.

4.C. THE INSTRUCTION BOOKLET SHALL BE ORGANIZED IN SECTIONS, WITH ONE SECTION PER SYSTEM. THE COVER OF THE INSTRUCTION BOOKLET SHALL BEAR THE NAME, ADDRESS AND PHONE NUMBER OF THE PROJECT, ARCHITECT, ENGINEER, MECHANICAL CONTRACTOR AND SUBCONTRACTORS.

4.D. REPRODUCIBLE "AS-BUILT" DRAWINGS INDICATING AS INSTALLED CONDITIONS SHALL BE PROVIDED TO THE ARCHITECT AFTER COMPLETION OF THE INSTALLATION.

4.E. PROVIDE "AS-BUILT" ON AUTOCAD 2010 OR LATER.

5. SHEET METAL WORK

5.A. EXCEPT AS OTHERWISE SHOWN OR NOTED, ALL DUCTWORK AND OTHER SHEET METAL WORK SHALL BE GALVANIZED SHEET STEEL AND SHALL BE INSTALLED IN ACCORDANCE WITH THE LATEST EDITION OF SHEET METAL AND AIR CONDITIONING CONTRACTORS NATIONAL ASSOCIATION, INC.

DUCTWORK STATIC PRESSURE CLASSIFICATIONS

PRESSURE CLASS	OPERATING CLASS
2 IN. W.G.	UP TO 2 IN. W.G.
6 IN. W.G.	UP TO 6 IN. W.G.

DUCTWORK SEALING REQUIREMENT SHALL BE SEAL CLASS

A.

5.B. VOLUME DAMPERS: GALVANIZED STEEL, PER SMACNA "LOW VELOCITY MANUAL," EXCEPT PROVIDE BEARING AT ONE END OF DAMPER ROD AND QUADRANT, WITH LEVER AND LOCKSCREW AT OTHER END. FOR INSULATED DUCTS, QUADRANTS MOUNTED ON COLLAR SHALL CLEAR INSULATION. INSTALL WITH ACCESSIBLE LEVERS.

5.C. ACCESS DOORS: INSULATED IF DUCT IS INSULATED; UNINSULATED IF DUCT IS UNINSULATED

5.C.1. PROVIDE MINIMUM 20 IN. X 14 IN. ON MAIN DUCTS, AND 12 IN. X 6 IN. ON BRANCH DUCTS, UNLESS OTHERWISE APPROVED, AT ALL DUCT ACCESSORIES (FIRE/SMOKE DAMPERS, LOUVERS, SMOKE DETECTORS, AUTOMATIC DAMPERS).

5.C.2. ALL ACCESS DOORS TO BE HINGED, WITH LATCH SIMILAR TO VENTLOCK NO. 100.

5.D. FLEXIBLE CONNECTIONS: NEOPRENE-COATED GLASS FABRIC WITH AN AREA DENSITY OF 30 OZ PER SQ YD. PROVIDE WITH SEWED AND CEMENTED SEAMS, SIMILAR TO VENT FABRICS AND WITH METAL COLLARS. ALLOW MINIMUM MOVEMENT OF 1 IN.

5.E. TURNING VANES: GALVANIZED STEEL SMALL DOUBLE-THICKNESS VANES WITH 2 IN. INSIDE RADIUS.

5.F. ALL DUCT DIMENSIONS INDICATED ON PLANS ARE INSIDE CLEAR DIMENSIONS.

5.G. AUTOMATIC DAMPERS: COMPLETE WITH LINKAGE AND ELECTRIC OPERATOR. OPPOSED BLADE DAMPER IS OF GALVANIZED STEEL, MIN. 4 IN. AND MAX. 8 IN. WIDE, WITH COMPRESSIBLE EDGE SEALS TO PREVENT LEAKAGE. FACTORY-ASSEMBLE STEEL LINKAGE AND SHAFT WITH NYLON OR OIL-IMPREGNATED BRONZE BEARINGS. MOTOR WITH SUFFICIENT POWER TO LIMIT LEAKAGE TO 10 CFM PER SQ FT. LINKAGE TO WITHSTAND LOAD EQUAL TO TWICE MAXIMUM OPERATING FORCE WITHOUT DEFLECTION. DAMPER MOUNTED IN WELDED STEEL CHANNEL FRAME.

5.H. EXTERIOR LOUVERS: 4" WIDE STATIONARY LOUVER MADE OF EXTRUDED ALUMINUM WITH 0.081" WALL THICKNESS, 6063TS ALLOY BLADE AND FRAME WITH STAINLESS STEEL OR ALUMINUM FASTENERS. LOUVER TO INCORPORATE STRUCTURAL SUPPORT TO WITHSTAND WIND LOAD OF 20 LBS PER SQ FT. PROVIDE REMOVABLE 3/4"x3/4" ALUMINUM BIRDSCREEN IN AN ALUMINUM FRAME. AIR PERFORMANCE AND WATER PENETRATION SHALL BE LESS THAN OR EQUAL TO RUSKIN MODEL ELF-375

5.I. COMBINATION FIRE AND SMOKE DAMPERS: UL LISTED GALVANIZED STEEL CONSTRUCTION AND OF MULTI-BLADED TYPE. THEY SHALL BE EQUIPPED WITH FIRESTAT OR RESETABLE LINK CONFORMING TO NFPA STANDARD 90A AND SIMILAR TO RUSKIN MODEL FSD 60. FIRE DAMPER SHALL CONFORM TO NFPA STANDARD 90A AND SIMILAR TO RUSKIN MODEL FD 60.

5.J. WIRED MESH SCREEN(WMS): NO. 16 USSG, 3/4 SQUARE MESH, IN 1" WIDE GALVANIZED STEEL ENCLOSING FRAME. FLANGED DUCT OPENING TO RECEIVE FRAME.

5.K. CHIMNEY AND BREECHINGS: IN ACCORDANCE WITH NFPA NO. 211. CLEAN OUT DOORS SHALL BE LOCATED AT CHANGES IN DIRECTION. DRAIN CAP SHALL BE PROVIDED AT THE BASE OF VERTICAL SECTION AND PIPED TO SUITABLE DRAINS. THE CHIMNEY AND BREECHING SHALL BE DOUBLE WALL TYPE, MATERIAL SHALL BE 304 STAINLESS STEEL.

6. AIR OUTLETS (ACCEPTABLE MANUFACTURERS: TITUS)

6.A. GENERAL

6.A.1. MARGIN TYPES, COLORS, FINISH AND METHODS OF ATTACHMENT FOR ALL DIFFUSERS, GRILLES AND REGISTERS SHALL BE COORDINATED WITH ARCHITECTURAL CEILING AND WALL DETAILS AND SPECIFICATIONS.

6.A.2. FRAME TYPE SHALL BE SUITABLE FOR MOUNTING IN CEILING OR WALL CONSTRUCTION AS INDICATED ON ARCHITECTURAL PLANS.

6.A.3. EXACT LOCATION OF ALL AIR OUTLETS SHALL BE PER ARCHITECTURAL PLANS.

6.A.4. SUITABLE FOR OPERATION AT 20% EXCESS TO 20% LESS THAN NOTED CAPACITY FOR CONSTANT VOLUME SYSTEMS AND AT 20% EXCESS TO 60% LESS THAN NOTED CAPACITY FOR VARIABLE VOLUME SYSTEMS. MANUFACTURER IS RESPONSIBLE FOR EXAMINING APPLICATION OF EACH OUTLET AND GUARANTEE THAT EACH WILL PROVIDE THE REQUIRED NC LEVELS AND COMFORT SPACE CONDITIONS WITHOUT DRAFTS THROUGHOUT OPERATING RANGE.

6.A.5. ALL REGISTERS AND DIFFUSERS SHALL BE PROVIDED WITH OPPOSED BLADE VOLUME DAMPERS. DAMPER OPERATING LEVERS SHALL BE ACCESSIBLE AT THE FACE OF AIR OUTLETS.

6.B. REGISTERS, DIFFUSERS AND GRILLES:

6.B.1. SUPPLY REGISTERS: ALUMINUM CONSTRUCTION AND ADJUSTABLE DOUBLE DEFLECTION ALUMINUM AIRFOIL LOUVERS, WITH VOLUME DAMPER. TITUS MODEL 300RL OR SIMILAR UNLESS OTHERWISE STATED.

6.B.2. NON-DUCTED TRANSFER EXHAUST AND RETURN GRILLES: STEEL CONSTRUCTION WITHOUT VOLUME DAMPER. TITUS PAR OR SIMILAR.

6.B.3. DUCTED TRANSFER AND EXHAUST GRILLES: STEEL CONSTRUCTION WITH VOLUME DAMPER. TITUS 350RL OR SIMILAR.

6.B.4. SQUARE CEILING SUPPLY DIFFUSERS: TITUS MODEL TMSA OR SIMILAR.

6.B.5. LINEAR SUPPLY DIFFUSER: EXTRUDED ALUMINUM CONSTRUCTION WITH AIR DEFLECTION VANE AND CABLE DAMPER IN EACH BRANCH TAP WITH 3 FEET CABLE TO DIFFUSER FACE. TITUS MODEL ML OR SIMILAR.

7. TESTING AND BALANCING

7.A. AIR BALANCING SHALL BE ACCOMPLISHED BY ADJUSTMENT OF FANS AND BRANCH DAMPERS FOR MAJOR ADJUSTMENTS. ADJUSTMENT OF TERMINAL DAMPERS AND DEVICES SHALL BE FOR TRIM OR MINOR ADJUSTMENT ONLY. THIS SHALL BE DONE TO PERMIT THE LEAST NOISE GENERATION IN THE TERMINAL AREAS AND UTILIZE MINIMUM FAN ENERGY.

7.B. WATER BALANCING SHALL BE ACCOMPLISHED BY ADJUSTMENT OF BALANCING VALVES AT PUMPS FOR PROPER FLOW. ADJUST FLOW THROUGH COILS AS REQUIRED.

7.C. UPON COMPLETION OF THE INSTALLATION, THE CONTRACTOR SHALL BALANCE AIR DISTRIBUTION SYSTEM AND WATER DISTRIBUTION SYSTEMS. THE CONTRACTOR TO TRAVERSE THE BASE BUILDING SUPPLY DUCT TO THE FLOORS AND SUPPLY DUCT FROM THE SUPPLEMENTAL AC UNITS.

7.D. THE CONTRACTOR SHALL PROVIDE ALL LABOR, PRESSURE GAUGES, FLOW METERS, SHEAVES, AND BELTS REQUIRED TO BALANCE SYSTEMS.

7.E. BALANCING REPORT SHALL BE PROVIDED ON AABC-TYPE FORMS.

7.F. FANS, AIR HANDLING UNITS, AND COILS SHALL BE BALANCED TO WITHIN +5% OF THEIR DESIGN CAPACITIES. ALL OTHER AIR AND WATER QUANTITIES SHALL BE BALANCED TO WITHIN +10% OF THE DESIGN QUANTITIES.

7.G. BALANCING AND TESTING SHALL BE PERFORMED AND SUPERVISED BY ONE OF THE FOLLOWING INDEPENDENT FIRMS SPECIALIZING IN TESTING AND BALANCING: (BUILDING ENGINEERS TO BE PRESENT DURING TESTING & BALANCING)

7.G.1. MERENDINO ASSOCIATES, INC.
7.G.2. ALL CITY TESTING AND BALANCING.
7.G.3. INDEPENDENT TESTING AND BALANCING

7.H. THE PERFORMANCE AND CAPACITY OF ALL SYSTEMS AND EQUIPMENT TO BE DEMONSTRATED BY THE CONTRACTOR.

7.I. LEAKAGE TEST TO BE COMPLETED ON KITCHEN EXHAUST DUCTWORK.

7.J. COPY OF BALANCING REPORT BY CERTIFIED AIR BALANCER SHALL BE PROVIDED TO OWNERSHIP

8. PIPE CLEANING AND TESTING

8.A. DURING CONSTRUCTION, PROPERLY PROTECT ALL OPEN ENDS OF PIPING TO PREVENT THE ENTRANCE OF SAND, DIRT, AND OTHER FOREIGN MATTER.

8.B. ALL PIPING SHALL BE HYDROSTATIC ALLY TESTED AT 1.5 TIMES THE MAXIMUM OPERATING PRESSURE(BUT NOT LESS THAN 100PSI) FOR 2 HOURS. PRESSURE TESTING MUST BE PERFORMED PRIOR TO THE INSTALLATION OF INSULATION.

8.C. BLOW DOWN OR CLEAN STRAINERS PERIODICALLY DURING THE CLEANING PROCESS.

8.D. PRIOR TO PROCEEDING WITH TIE-INS SECURE BLEED-OFF LINES, DISCONNECT ANY TEMPORARY EQUIPMENT AND OBTAIN APPROVAL FROM THE OWNER.

8.E. THE WATER TREATMENT SUPPLIER SHALL CERTIFY THE ACCURACY OF THE CLEANOUT/FLUSH, PASSIVATING AND CORROSION INHIBITING PROCEDURES PERFORMED UNDER HIS SUPERVISION.

9. INSULATION - GENERAL REQUIREMENTS

9.A. ALL INSULATION MATERIALS, INCLUDING JACKETS, FACING, ADHESIVE, COATINGS, AND ACCESSORIES ARE TO BE FIRE HAZARD RATED AND LISTED BY UNDERWRITERS LABORATORIES, INC. USING STEINER TUNNEL TEST METHOD FOR FIRE HAZARD CLASSIFICATION OF BUILDING MATERIALS, STANDARD UL 723 (ASTM E-84). (ASA A2.5-1963). FLAMESPREAD: MAXIMUM 25. FUEL CONTRIBUTED AND SMOKE DEVELOPED: MAXIMUM 50. FLAMEPROOFING TREATMENTS SUBJECT TO DETERIORATION FROM MOISTURE OR HUMIDITY ARE NOT ACCEPTABLE.

9.B. DEFINITIONS:

9.B.1. CONCEALED: INDOOR DUCTS, PIPING OR EQUIPMENT WHICH ARE NOT EXPOSED.

9.B.2. EXPOSED: INDOOR DUCTS, PIPING OR EQUIPMENT LOCATED IN MECHANICAL EQUIPMENT ROOMS AND IN AREAS WHICH WILL BE VISIBLE WITHOUT REMOVING CEILING OR OPENING ACCESS PANELS.

9.B.3. OUTDOOR: DUCT, PIPING OR EQUIPMENT WHICH IS EXPOSED TO THE WEATHER.

10. PIPING INSULATION

10.A. INSULATE ALL PIPING IN ACCORDANCE WITH INSULATION SCHEDULE EXCEPT AS OTHERWISE NOTED. ALL PIPING SHALL BE LABELED FOR IDENTIFYING THE SYSTEMS.

INSULATION SCHEDULE - PIPING

SERVICE	LOW TEMP. PIPE (40-100°F)		LOW TEMP. FITTING AND VALVES (40-100°F)		HIGH TEMP. PIPE (100-250°F)		HIGH TEMP. FITTINGS AND VALVES (100-250°F)
SIZE	1.5">	1.5"≤	1.5">	1.5"≤	1.5">	1.5"≤	1.5">
THICKNESS	1.5"	2"	1.5"	2"	1.5"	2"	1.5"
MATERIAL	P-1		P-4		P-1		P-4
FINISH	VAPORSEAL		VAPORSEAL F-1		-		F-1

10.B. PIPING, VALVES, AND FITTINGS TO BE INSULATED:

10.B.1. LOW TEMPERATURE PIPING SYSTEMS - 40 TO 100°F INCLUDING:
10.B.1.a. CONDENSER WATER SUPPLY AND RETURN
10.B.1.b. CONDENSATE DRAIN PIPING

10.B.2. HOT WATER PIPING SYSTEMS - 100 TO 250°F INCLUDING:
10.B.2.1. HOT WATER SUPPLY AND RETURN.

10.C. MATERIAL:

10.C.1. TYPE P-1: MIN. 4 LB DENSITY MOLDED FIBERGLASS W/ MAX. 0.23 K-FACTOR AT 75°F MEAN TEMPERATURE, FACTORY-APPLIED FIRE-RETARDANT FOIL-SKRIM-KRAFT FACING, AND ALL SERVICE JACKET. OWENS-CORNING 650 ASJ OR SIMILAR.

10.C.2. TYPE P-3: MIN. 4 LB DENSITY MOLDED FIBERGLASS FITTING W/ MAX. 0.23 K-FACTOR AT 75°F MEAN TEMPERATURE. EPOLUX HAMFAB MOLDED FITTINGS OR SIMILAR.

10.C.3. TYPE P-4: MIN. 1 LB DENSITY FIBERGLASS FITTING INSERTS W/ MAX. 0.28 K-FACTOR AT 75°F MEAN TEMPERATURE. MANVILLE HI-LO TEMP INSULATION INSERTS OR SIMILAR.

10.C.4. TYPE P-6: MIN. 6 LB MOLDED FOAMED PLASTIC W/ MAX. 0.27 K-FACTOR AT 75°F MEAN TEMPERATURE AND MAX. 0.17 PERMEANCE. ARMSTRONG ARMAFLEX II OR SIMILAR.

10.D. FINISH:

10.D.1. TYPE F-1: FITTING COVER W/ MOLDED WHITE PVC JACKET, UL CLASS 1, AND MAXIMUM PERMEANCE OF 0.05. MANVILLE ZESTRON OR SIMILAR.

10.D.2. TYPE F-2: WHITE VAPOR BARRIER COATING W/ 10X10 OR 20X20 MESH WHITE GLASS, POLYESTER OR NYLON CLOTH REINFORCING MEMBRANE, MIN. 31 MM DRY FILM THICKNESS AND UL CLASSIFIED. FOSTER TITE-FIT OR SIMILAR.

10.D.3. TYPE F-4: ALUMINUM JACKETING W/ MINIMUM 0.016 IN. WALL THICKNESS AND LONGITUDINAL JOINTS WITH LOCK SEAMS.

10.D.4. TYPE F-6: WHITE FINISHING AND INSULATING CEMENT APPLIED OVER HEXAGONAL WIRE MESH. KEENE SUPERSLICK OR SIMILAR

10.E. INSTALLATION:

10.E.1. BEFORE APPLYING INSULATION ALL PRESSURE AND LEAK TESTS SHALL BE COMPLETED AND APPROVED.

10.E.2. ALL INSULATION SHALL BE BUTTED FIRMLY TOGETHER. PROVIDE 2 IN. LAMP STRIPS AT ALL SEAMS SECURED WITH ADHESIVE. USE VAPOR BARRIER TAPE AND VAPORSEAL ADHESIVE WHERE REQUIRED. STAPLES NOT PERMITTED. REFRIGERANT PIPING INSULATION SHALL HAVE MITERED FITTINGS.

10.E.3. ALL INSULATION AND VAPOR BARRIERS SHALL BE CONTINUOUS PASSING THROUGH SLEEVES, HANGERS, ETC., OR OTHER OPENINGS. PROVIDE SADDLES OR SHIELDS FOR PROTECTION.

10.E.4. INSULATION FOR STRAINERS OR OTHER FITTINGS OR ACCESSORIES REQUIRING SERVICING OR INSPECTION SHALL HAVE INSULATION REMOVABLE AND REPLACEABLE WITHOUT DAMAGE.

11. DUCTWORK INSULATION:

11.A. INSULATE ALL DUCTWORK IN ACCORDANCE WITH INSULATION SCHEDULE EXCEPT AS OTHERWISE NOTED.

INSULATION SCHEDULE

SERVICE	SUPPLY	OAI	SUPPLY/RETURN	EXHAUST
TYPE	CONCEALED	ALL	EXPOSED	MER EXPOSED
THICKNESS	1"	2"	1"	2"
MATERIAL	D-1	D-3	D-2	D-3
FINISH	VAPORSEAL	VAPORSEAL	VAPORSEAL	VAPORSEAL

11.B. REINSULATE ALL DUCTWORK AND PIPING WHICH IS EXISTING AND DAMAGED DURING CONSTRUCTION OR SHOWN OR REQUIRED TO BE RELOCATED. INSULATE WITH SAME MATERIAL AND THICKNESS.

11.C. NON-INSULATED DUCTWORK:

11.C.1. WHERE SOUNDLINING IS OF MINIMUM THICKNESS SPECIFIED FOR INSULATION.

11.C.2. AIR CONDITIONING RETURN AIR DUCTWORK EXPOSED IN AIR CONDITIONED SPACES AND INSTALLED IN HUNG CEILINGS WHERE SPACE IMMEDIATELY ABOVE AND BELOW ARE BOTH AIR CONDITIONED.

11.D. MATERIAL:

11.D.1. TYPE D-1: MIN. 1-LB DENSITY FIBERGLASS BLANKET W/ MAX. 0.28 K-FACTOR AT 75° MEAN TEMPERATURE AND FACTORY-APPLIED FOIL- SKRIM-KRAFT FACING. MANVILLE MICROLITE OR SIMILAR.

11.D.2. TYPE D-2: 3 LB FIBERGLASS BOARD W/ MAX. K-FACTOR SHALL BE 0.23 AT 75°F MEAN TEMPERATURE WITH A MINIMUM DENSITY OF 3-LB. THE INSULATION SHALL BE PROVIDED WITH A FACTORY-APPLIED ALL PURPOSE OR ALL SERVICE FACING. THE INSULATION SHALL BE EQUAL TO MANVILLE TYPE 814 SPIN-GLAS AP.

11.D.3. TYPE D-3: MINIMUM 6 LB FIBERGLASS BOARD. MAXIMUM 0.22 K-FACTOR AT 75° MEAN TEMPERATURE WITH FACTORY APPLIED ALL PURPOSE OR ALL SERVICE FACING. SIMILAR TO MANVILLE 817 SPIN-GLAS AP.

11.E. INSTALLATION

11.E.1. FIBERGLASS BLANKET: 2 INCH LAP STRIPS AT ALL SEAMS. SECURE BOTTOM OF ALL DUCTS OVER 24 INCH WIDE WITH MINIMUM 2 ROWS OF WELD PINS 12 INCH O CENTER. SECURE ALL SEAMS WITH FOIL VAPOR BARRIER TAPE AND VAPORSEAL ADHESIVE.

11.E.2. FIBERGLASS BOARD: SEAL JOINTS AND BREAKS IN FACING WITH 3 INCH WIDE TAPE TO MATCH FACING AND ADHERE WITH VAPORSEAL ADHESIVE. APPLY 5 INCH WIDE TAPE AT CORNERS, WELD PINS ON TOP, SIDES AND BOTTOM.

12. VIBRATION ISOLATION

12.A. GENERAL

12.A.1. PROVIDE ISOLATION FOR EQUIPMENT, PIPING AND DUCTWORK.

12.A.2. INSTALL IN ACCORDANCE WITH MANUFACTURER'S INSTRUCTIONS.

12.A.3. PROVIDE LEVELING DEVICES AND APPROVED RESILIENT RESTRAINING DEVICES AS REQUIRED TO LIMIT EQUIPMENT AND PIPING MOTION IN EXCESS OF 1/4 IN.

12.A.4. ACCEPTABLE MANUFACTURERS:
12.A.4.a. MASON INDUSTRIES, INC.
12.A.4.b. KORFUND DYNAMICS CORP.
12.A.4.c. VIBRATION ELIMINATOR CO.

12.B. CEILING-HUNG FANS AND EQUIPMENT

12.B.1. ALL SUSPENDED AC UNITS SHALL BE HUNG FROM COMBINATION SPRING/ NEOPRENE HANGERS SUCH AS MASON INDUSTRIES TYPE 30N HANGERS HAVING A MINIMUM 1.25" STATIC DEFLECTION.

12.B.2. ALL FANS SHALL BE HUNG FROM MASON TYPE HD HANGERS HAVING A MINIMUM STATIC DEFLECTION OF 0.35".

12.B.3. PROVIDE SUPPLEMENTAL STEEL AS REQUIRED WHERE EQUIPMENT OR STRUCTURE CANNOT SUPPORT POINT LOADS.

13. PIPING SYSTEMS:

13.A. PIPING LAYOUT AS SHOWN IS A SCHEMATIC REPRESENTATION OF THE DESIGN INTENT. THE CONTRACTOR SHALL BE RESPONSIBLE FOR THE PROPER ERECTION OF A PIPING SYSTEM SUITABLE FOR THE INTENDED WORK.

13.B. ALL PIPING SHALL BE CONCEALED WHENEVER POSSIBLE. EXPOSED DUCTWORK SHALL RUN PERPENDICULAR AN/OR PARALLEL TO FLOORS, WALLS, ETC.

13.C. PIPING MATERIALS / PIPE FITTINGS

SERVICE	SIZE	MATERIAL	FITTING
HOT WATER	3.5">	COPPER TYPE L HARD TEMPER (DRAW) IN ACCORDANCE WITH ASTM STANDARD SPEC. B-